

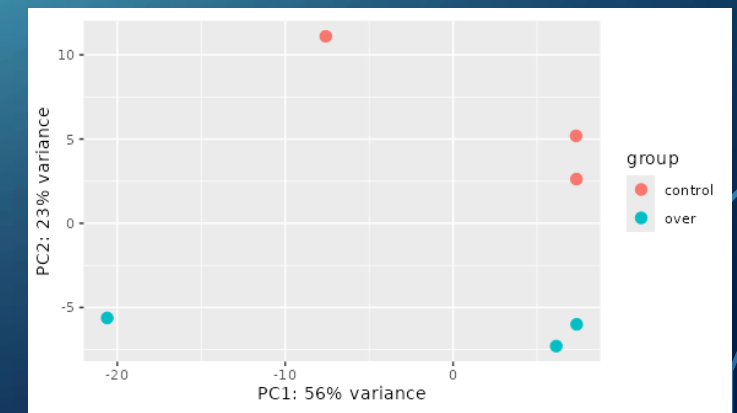
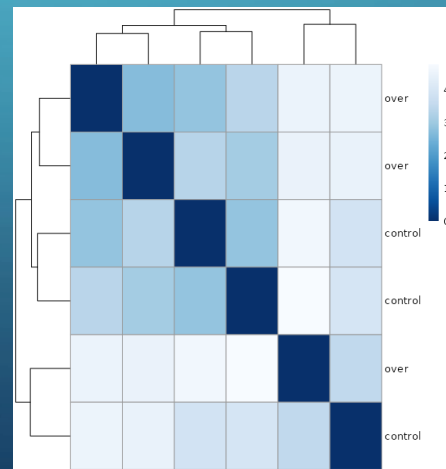
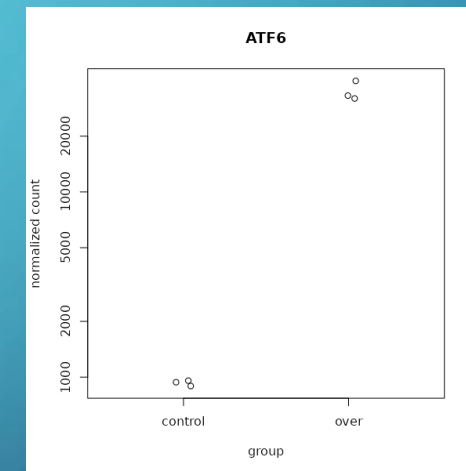
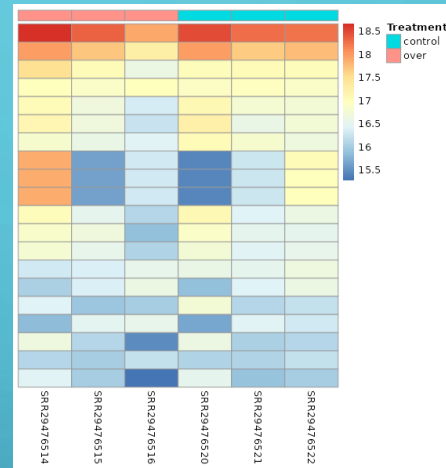


NGS SEQUENCING ANALYSIS

RNA-SEQ: PART 3

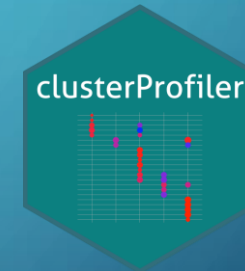
RECAP

- We performed differential expression (DE) analysis on the processed RNA-seq data using the DESeq2 R package
- We created some preliminary graphs, such as a heatmap, distance matrix, and principal component analysis (PCA) plot



TODAY

- Perform in-depth analysis on the DE results
- Recreate figures from the article



OPEN ACCESS PEER-REVIEWED

RESEARCH ARTICLE

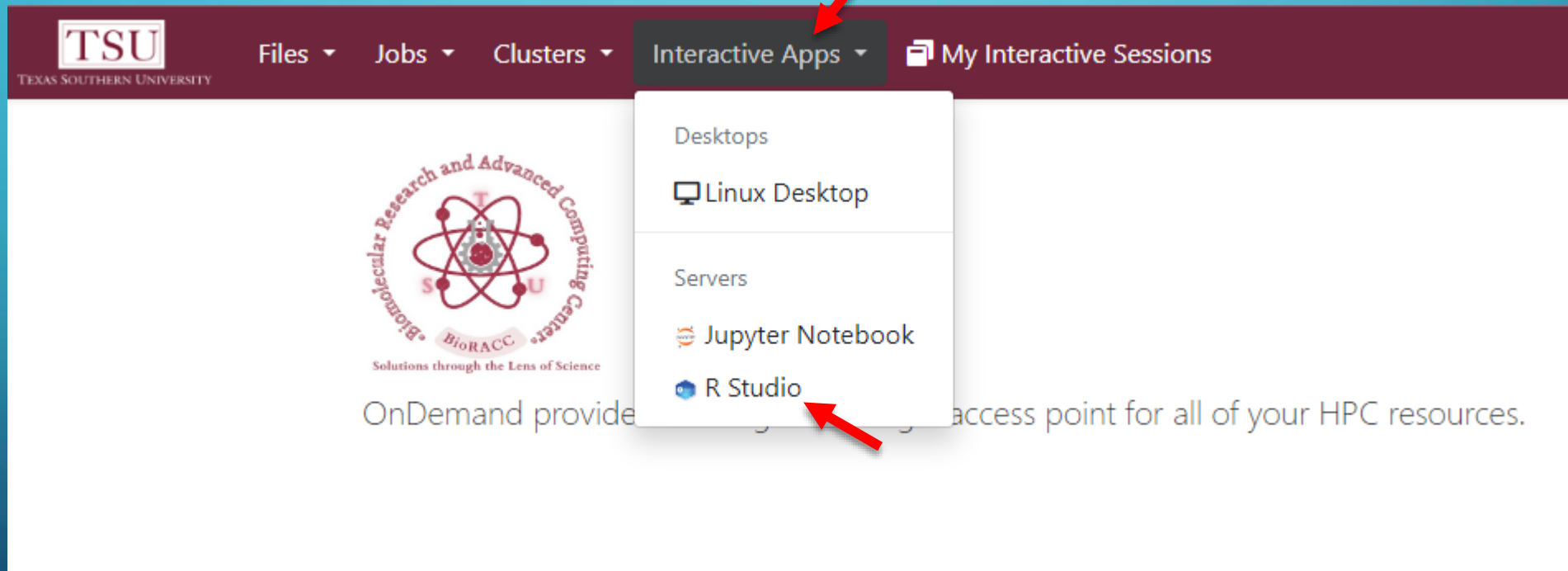
Transcriptomic analysis of cellular senescence induced by ectopic expression of ATF6α in human breast cancer cells

Ju Won Kim , So-Hyun Bae , Yesol Moon, Eun Kyung Kim, Yongjin Kim, Yun Gyu Park, Mi-Ryung Han , Jeongwon Sohn 

Published: October 28, 2024 • <https://doi.org/10.1371/journal.pone.0309749>

REQUESTING AN RSTUDIO SESSION

Interactive Apps → R Studio



REQUESTING AN RSTUDIO SESSION

Select **4 Cores** and enter **2** for the number of hours, then click Launch

TSU
TEXAS SOUTHERN UNIVERSITY

Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾

Home / My Interactive Sessions / R Studio

Interactive Apps

- Desktops
- Linux Desktop
- Servers
- Jupyter Notebook
- R Studio**

R Studio

How many resources do you need for this desktop?

☒ 4 Cores

☐ 8 Cores

☐ 16 Cores, 1 GPU

☐ 32 Cores, 1 GPU

How many hours do you need the resources for?

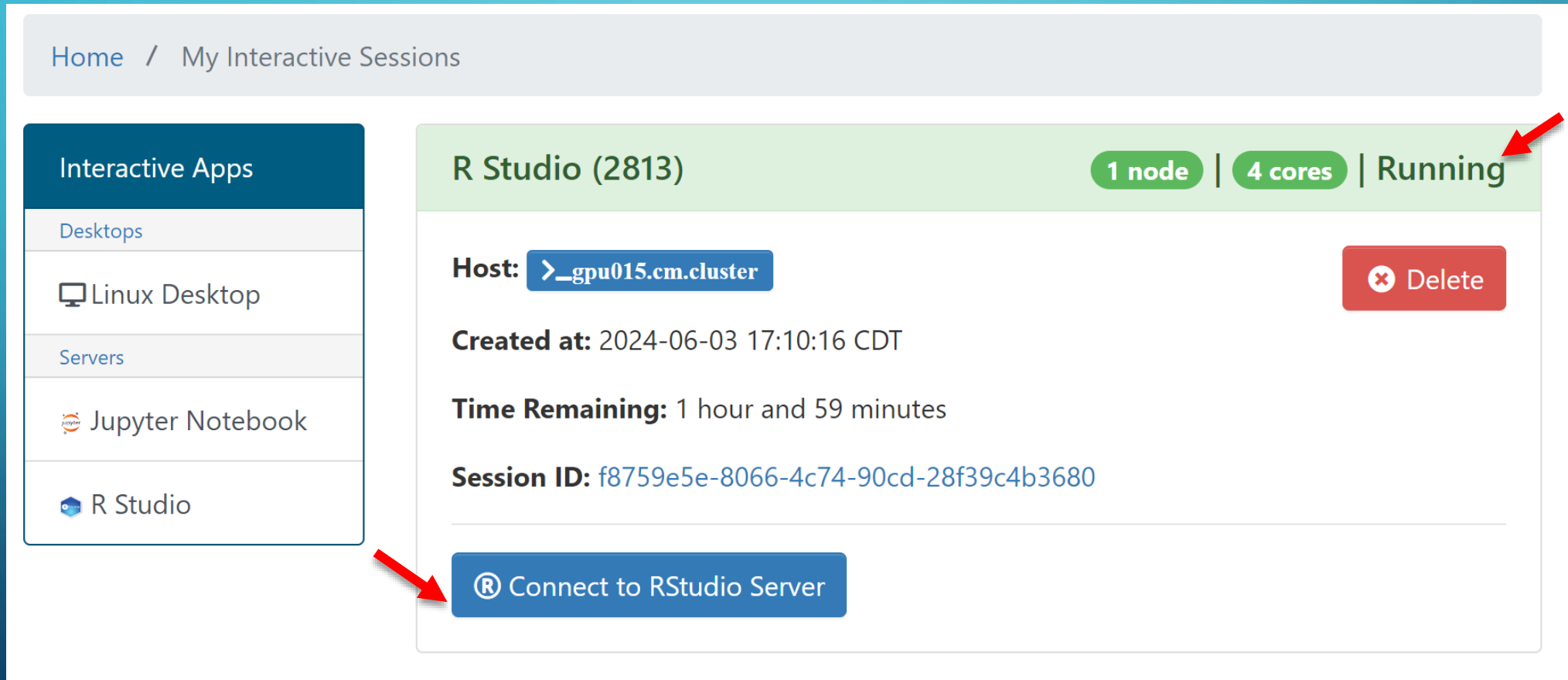
2

Launch

* The R Studio session data for this session can be accessed under the [data root](#) directory.

REQUESTING AN RSTUDIO SESSION

Click *Connect to Rstudio Server* when session is running



The screenshot displays a web interface for managing interactive sessions. On the left, a sidebar titled 'Interactive Apps' contains a list of available applications: 'Desktops', 'Linux Desktop', 'Servers', 'Jupyter Notebook', and 'R Studio'. The 'R Studio' option is selected. The main content area shows details for an active session named 'R Studio (2813)'. The session status is 'Running', with '1 node' and '4 cores' allocated. A red arrow points to the 'Running' status. Below the session name, the 'Host' is listed as '>_gpu015.cm.cluster', the 'Created at' time is '2024-06-03 17:10:16 CDT', the 'Time Remaining' is '1 hour and 59 minutes', and the 'Session ID' is 'f8759e5e-8066-4c74-90cd-28f39c4b3680'. A red 'Delete' button is located next to the host information. At the bottom of the session details, a blue button labeled '® Connect to RStudio Server' is highlighted with a red arrow.

Home / My Interactive Sessions

Interactive Apps

- Desktops
- Linux Desktop
- Servers
- Jupyter Notebook
- R Studio

R Studio (2813) 1 node | 4 cores | Running

Host: >_gpu015.cm.cluster Delete

Created at: 2024-06-03 17:10:16 CDT

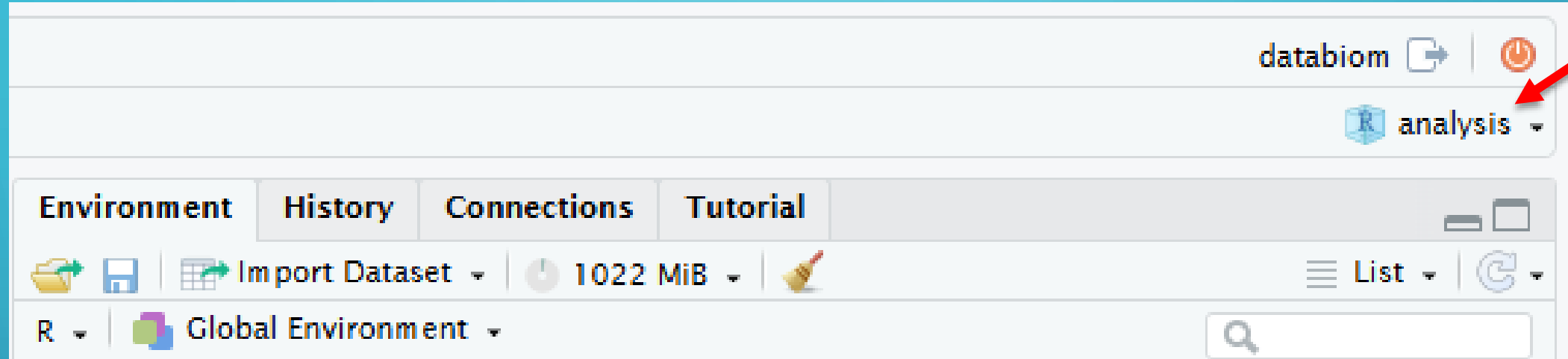
Time Remaining: 1 hour and 59 minutes

Session ID: f8759e5e-8066-4c74-90cd-28f39c4b3680

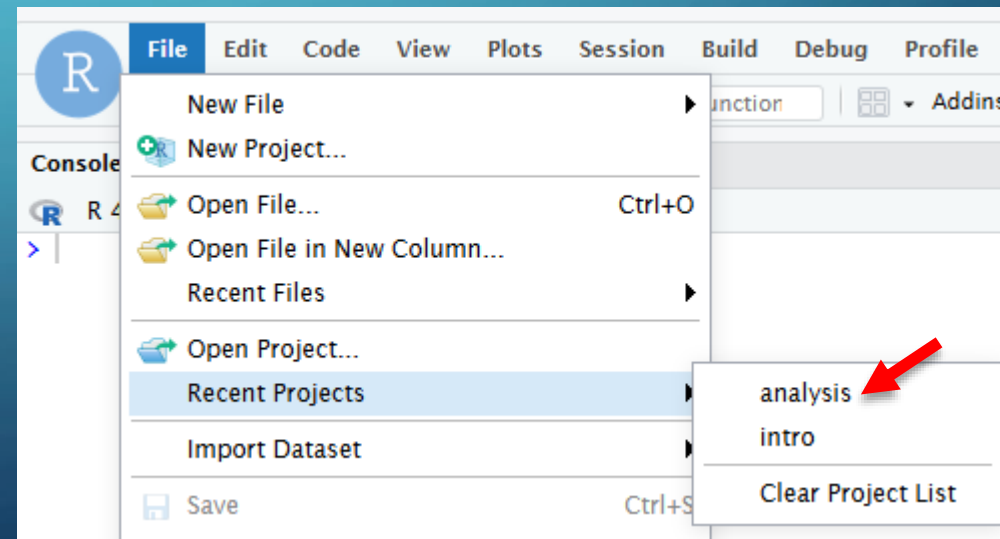
® Connect to RStudio Server

RSTUDIO

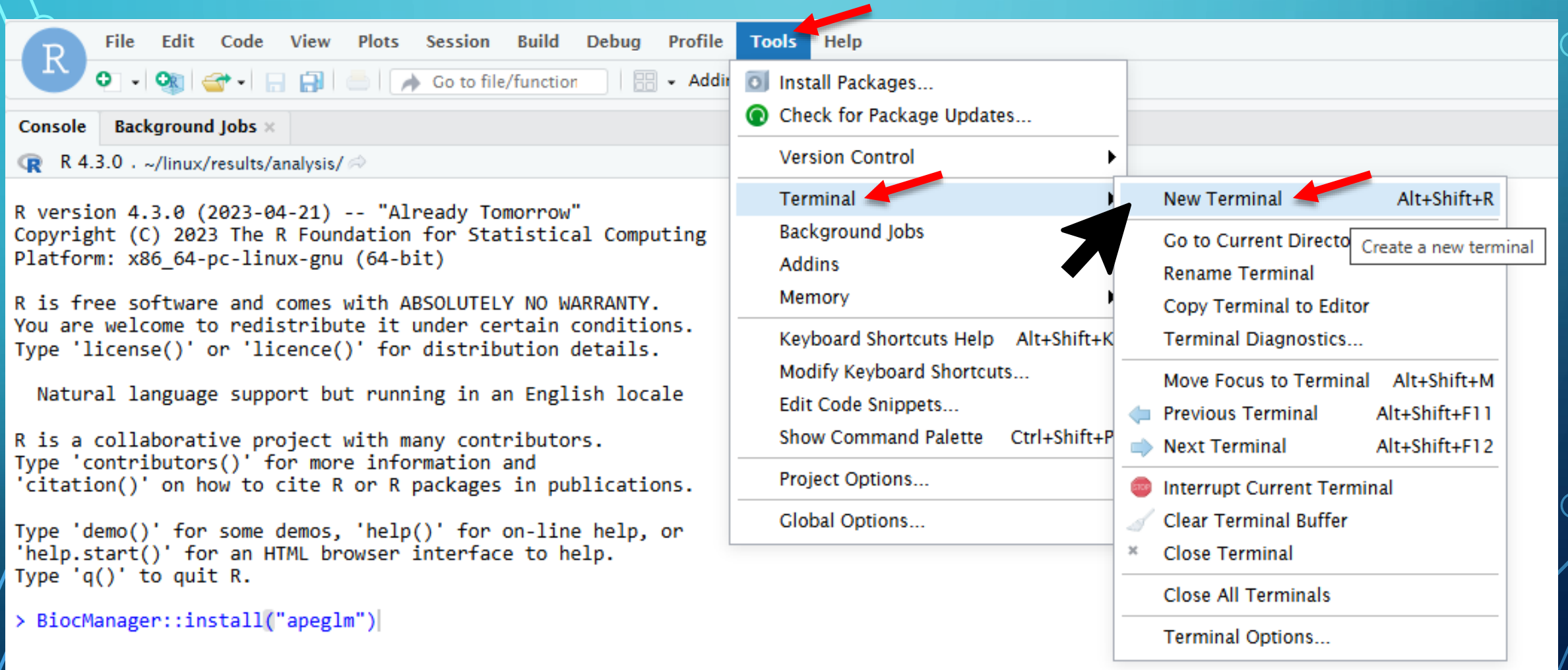
Your 'analysis' project should automatically be loaded



*If not, load the 'analysis.Rproj' project:



OPEN A TERMINAL WITHIN RSTUDIO



COPY AN R SCRIPT TO YOUR DIRECTORY

Console

Terminal x

Background Jobs x



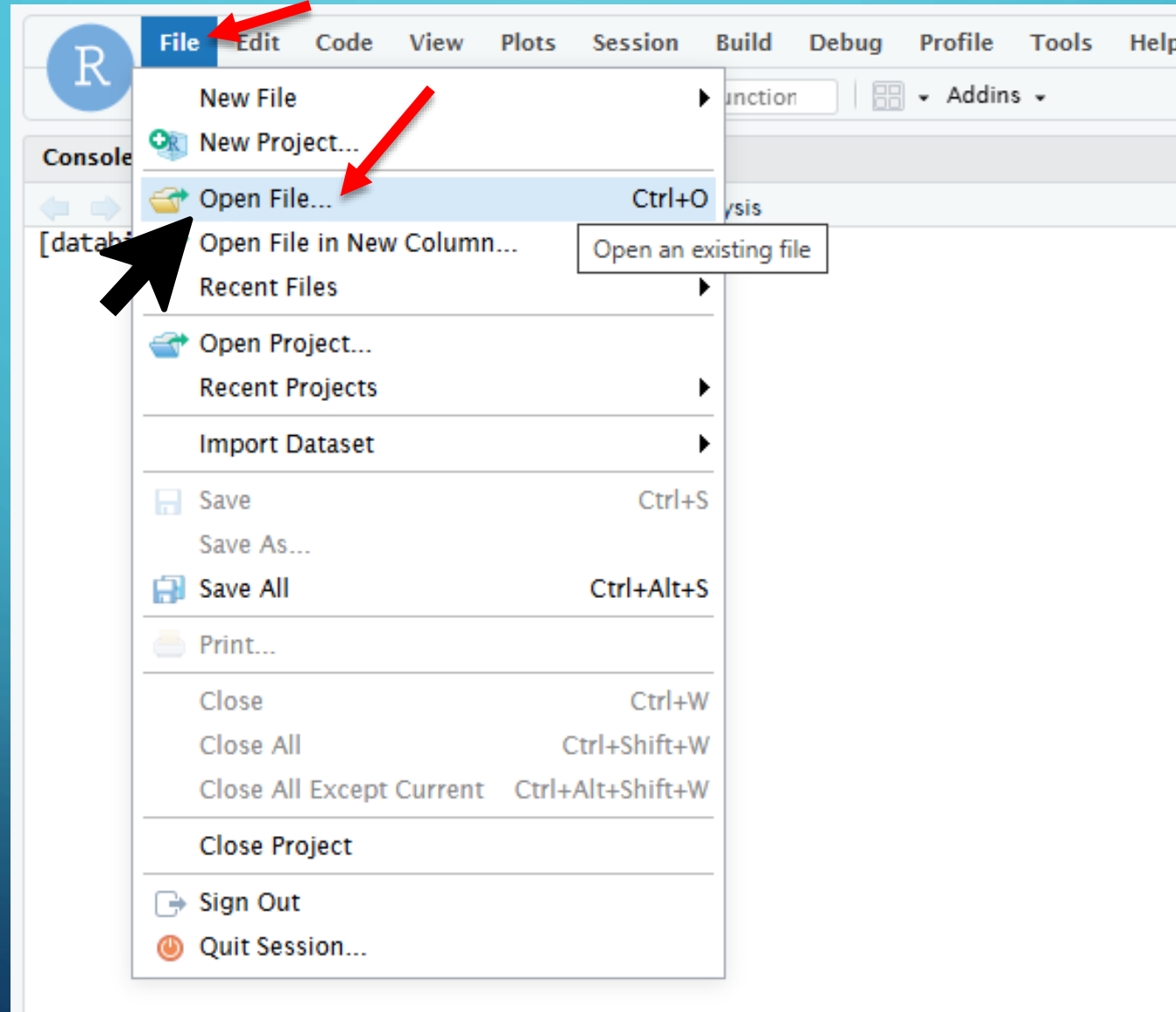
Terminal 1 ▾

/home/databiom/linux/results/analysis

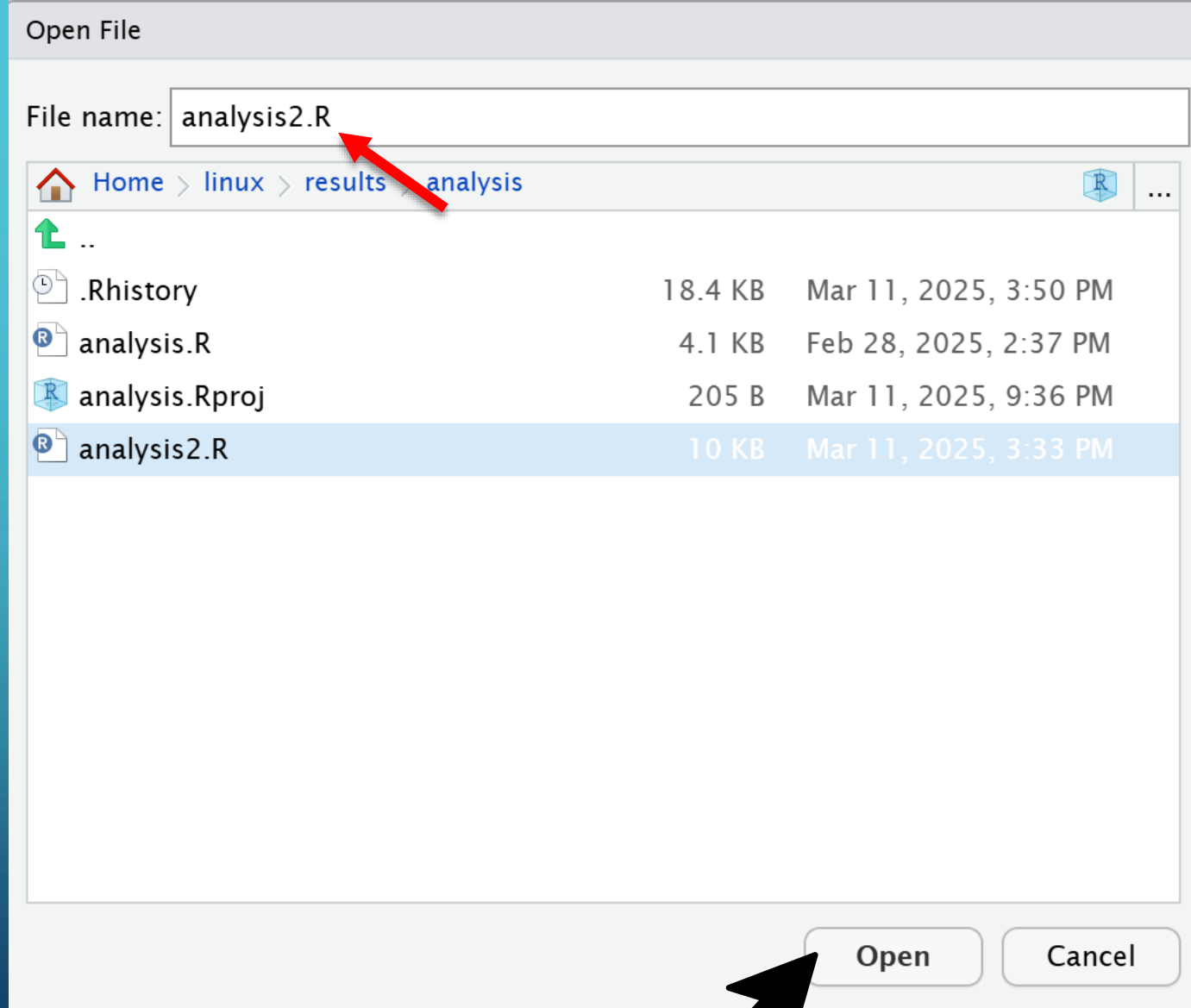
```
[databiom@gpu001 analysis]$ cp /home/databiom/analysis2.R .
```

Press "enter"

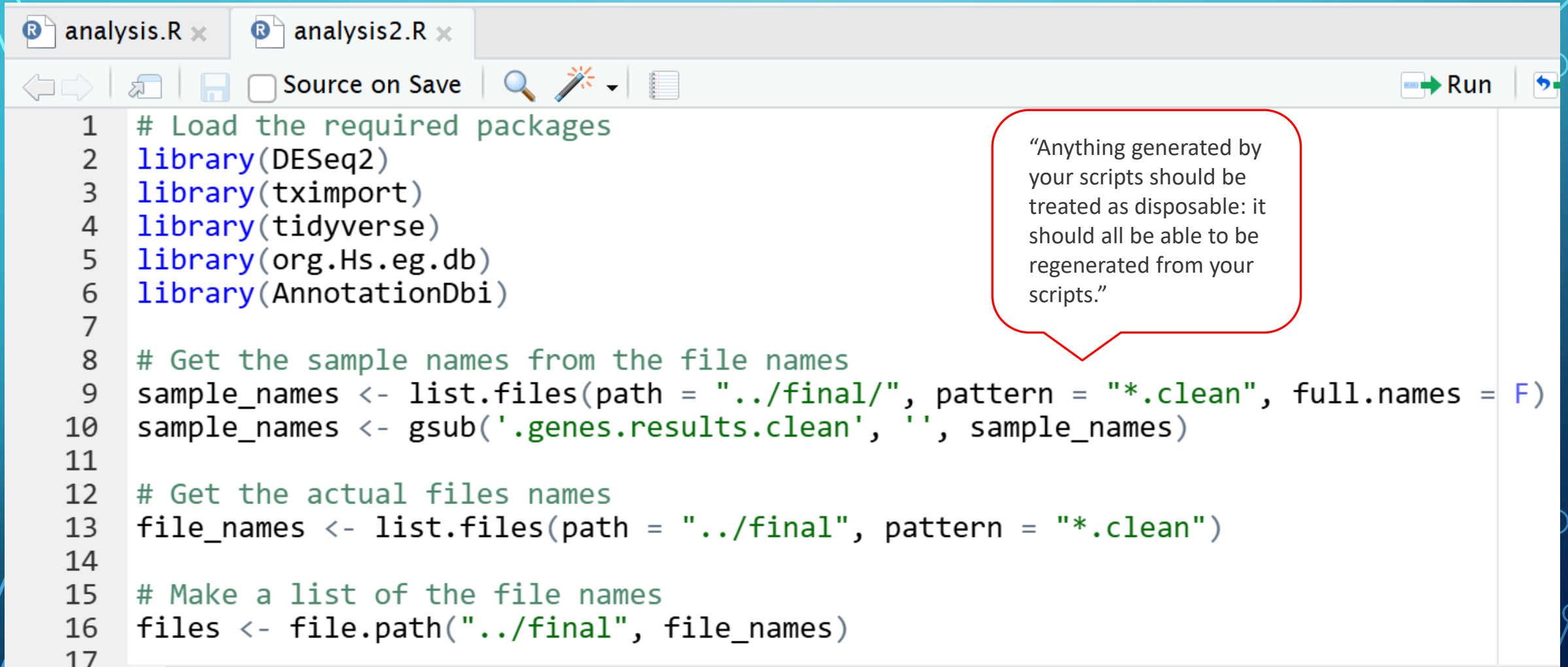
OPEN THE SCRIPT



OPEN THE SCRIPT



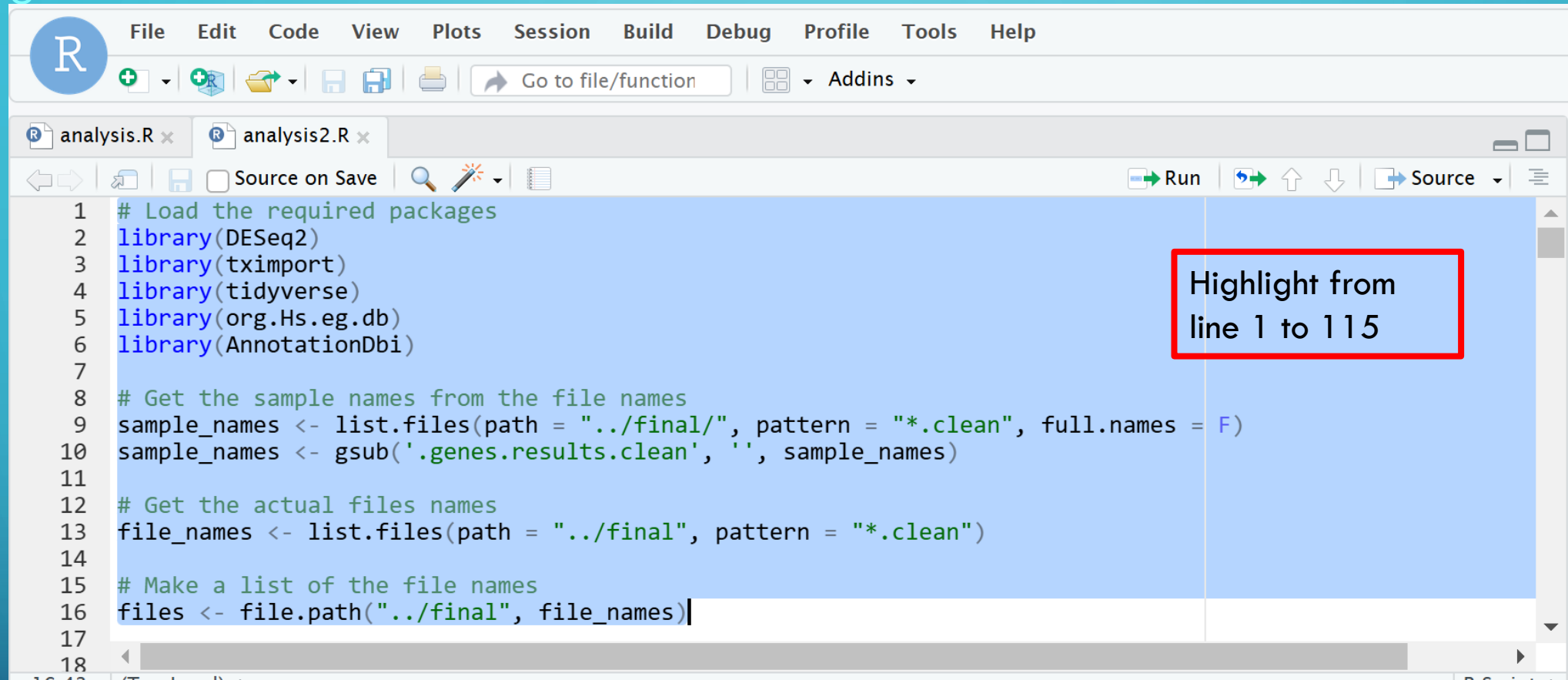
ANALYSIS2.R SCRIPT



```
analysis.R x analysis2.R x
Source on Save Run
1 # Load the required packages
2 library(DESeq2)
3 library(tximport)
4 library(tidyverse)
5 library(org.Hs.eg.db)
6 library(AnnotationDbi)
7
8 # Get the sample names from the file names
9 sample_names <- list.files(path = "../final/", pattern = "*.clean", full.names = F)
10 sample_names <- gsub('.genes.results.clean', '', sample_names)
11
12 # Get the actual files names
13 file_names <- list.files(path = "../final", pattern = "*.clean")
14
15 # Make a list of the file names
16 files <- file.path("../final", file_names)
17
```

"Anything generated by your scripts should be treated as disposable: it should all be able to be regenerated from your scripts."

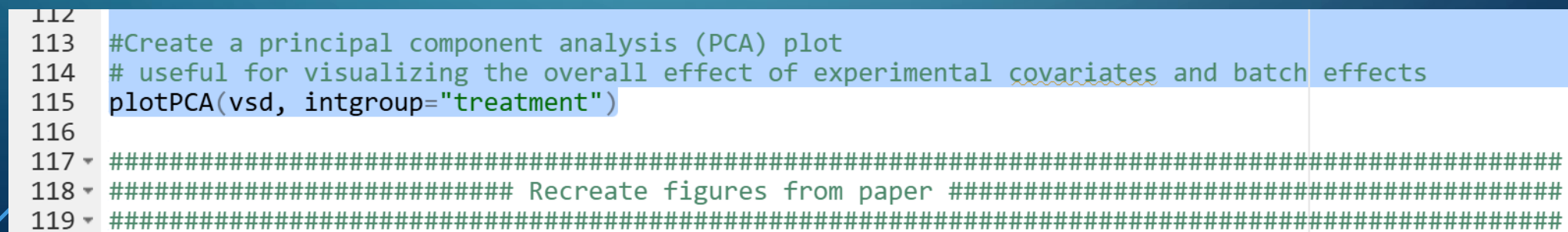
HIGHLIGHT PREVIOUS LINES OF CODE



The screenshot shows the RStudio IDE interface. The menu bar includes File, Edit, Code, View, Plots, Session, Build, Debug, Profile, Tools, and Help. The toolbar shows icons for file operations and a search bar. The script editor displays the following code:

```
1 # Load the required packages
2 library(DESeq2)
3 library(tximport)
4 library(tidyverse)
5 library(org.Hs.eg.db)
6 library(AnnotationDbi)
7
8 # Get the sample names from the file names
9 sample_names <- list.files(path = "../final/", pattern = "*.clean", full.names = F)
10 sample_names <- gsub('.genes.results.clean', '', sample_names)
11
12 # Get the actual files names
13 file_names <- list.files(path = "../final", pattern = "*.clean")
14
15 # Make a list of the file names
16 files <- file.path("../final", file_names)
```

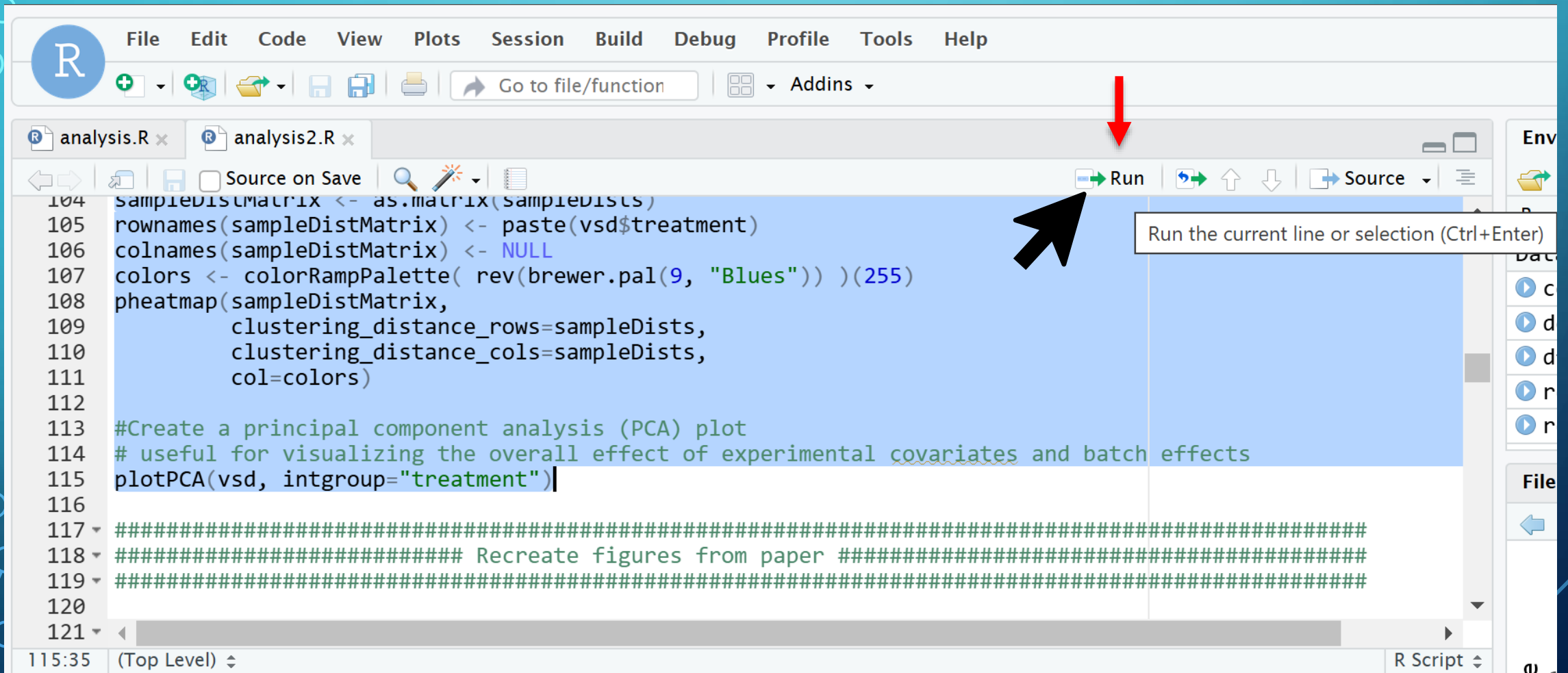
A red box highlights the text "Highlight from line 1 to 115" in the right margin.



The continuation of the R script shows the following code:

```
112
113 #Create a principal component analysis (PCA) plot
114 # useful for visualizing the overall effect of experimental covariates and batch effects
115 plotPCA(vsd, intgroup="treatment")
116
117 #####
118 ##### Recreate figures from paper #####
119 #####
```

RUN PREVIOUS LINES OF CODE



RECREATE FIGURES: FIG. 1B

Complex Heatmap

Here, we filter our results for adjusted p-value < 0.05 and absolute $\log_2$ fold change > 1 .

```
121 ▾ ##### Heatmap for 9hr treatments # adjusted p-value < 0.05 and |log2fc|> 1 ##### Fig. 1B #####
122 # Filter the results for adjusted p-value less than 0.05 and absolute value of log2 fold change greater than 1
123 resLFC.05fc1 <- as.data.frame(subset(resLFC, padj < 0.05 & abs(log2FoldChange)>1))
124 # Check # of genes left (65)
125 nrow(resLFC.05fc1)
126 # Convert the VST transformed data into a data frame
127 vsd_df <- as.data.frame(assay(vsd))
128 # Filter the matrix based on above criteria for adjusted p-value and log2fc
129 vsd_df <- vsd_df[row.names(vsd_df) %in% row.names(resLFC.05fc1),]
130 # Check if filtered genes is correct (should be 65)
131 nrow(vsd_df)
```



To run code, either click the 'Run'  bottom at the top-right of the script or use the ctrl+Enter shortcut. Wherever your cursor is, the line(s) of code will be executed.

RECREATE FIGURES: FIG. 1 B

Complex Heatmap

Use gene symbols for labeling rows and convert to a matrix:

```
132 # Add symbols to the matrix
133 vsd_df$symbol <- mapIds(org.Hs.eg.db, keys = row.names(vsd_df),
134                        column = "SYMBOL",
135                        keytype = "ENSEMBL",
136                        multiVals = "first")
137 # Look at symbols - some are NA - these are long non-coding RNAs (lncRNAs) that don't have official gene symbols
138 vsd_df$symbol
139 # create a vector of gene symbols to use for the heatmap row labels
140 row_labels <- vsd_df$symbol
141 # remove the 'symbol' column from the data frame (need to do to make a matrix of just numbers)
142 vsd_df2 <- subset(vsd_df, select=-(symbol))
143 # Convert the data frame to a matrix
144 vsd_matrix <- as.matrix(vsd_df2)
```

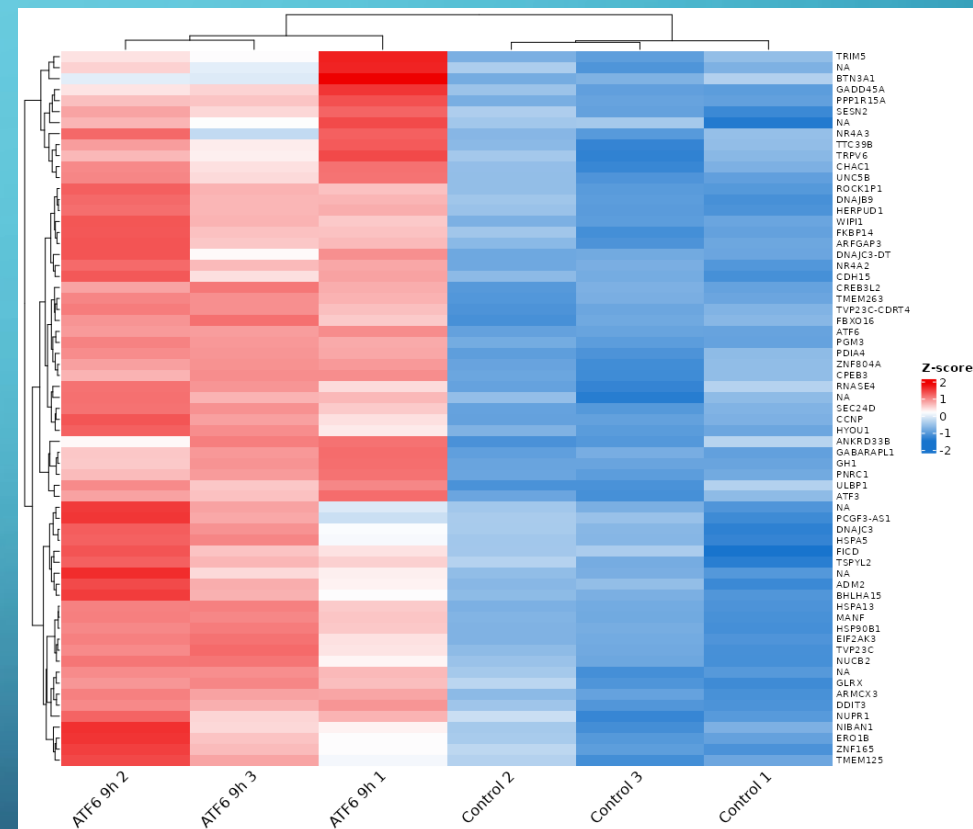
RECREATE FIGURES: FIG. 1 B

ComplexHeatmap

Scale values, change column names, create a color palette, and create heatmap:

```
145 # Scale (Z-score) the gene counts for better visualization
146 vsd_scaled <- t(scale(t(vsd_matrix)))
147 # Change column names to be more informative (*need to know order*)
148 colnames(vsd_scaled) <- c("ATF6 9h 1", "ATF6 9h 2", "ATF6 9h 3", "Control 1", "Control 2", "Control 3")
149 # Create a color palette
150 myCol <- colorRampPalette(c("dodgerblue3", "white", "red2"))(n = 100)
151 # Load packages
152 library(ComplexHeatmap)
153 library(circlize)
154 library(colorRamp2)
155 # Create heatmap
156 Heatmap(vsd_scaled, col = myCol, column_names_rot=45, row_labels = row_labels,
157         row_names_gp = gpar(fontsize = 8), row_names_max_width = unit(3, "cm"),
158         heatmap_legend_param = list(title = "Z-score"))
```

Complex Heatmap



RECREATE FIGURES: FIG. 1D

Here, we add symbols to the shrunken $\log_2$ fold change results and use the default settings of the EnhancedVolcano package:

```
161 ▾ ##### Volcano Plot ##### Fig. 1D #####
162 # Add symbols to the adjusted log2fc data
163 resLFC$symbol <- mapIds(org.Hs.eg.db, keys = row.names(res),
164                        column = "SYMBOL",
165                        keytype = "ENSEMBL",
166                        multiVals = "first")
167 # Enhanced Volcano plot # default settings
168 library(EnhancedVolcano)
169 EnhancedVolcano(resLFC,
170                lab = resLFC$symbol,
171                x = 'log2FoldChange',
172                y = 'padj')
```

FIG. 1D

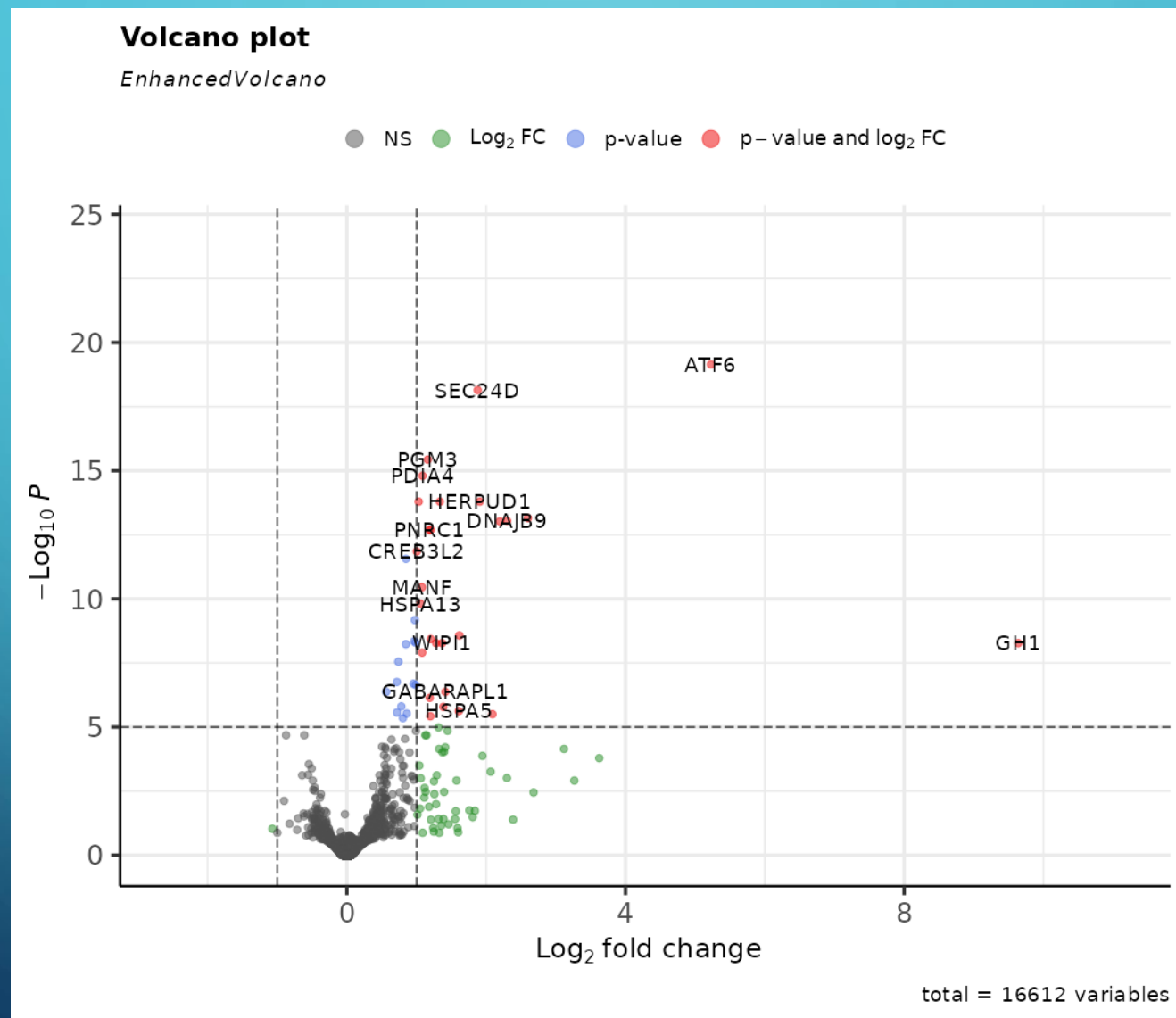
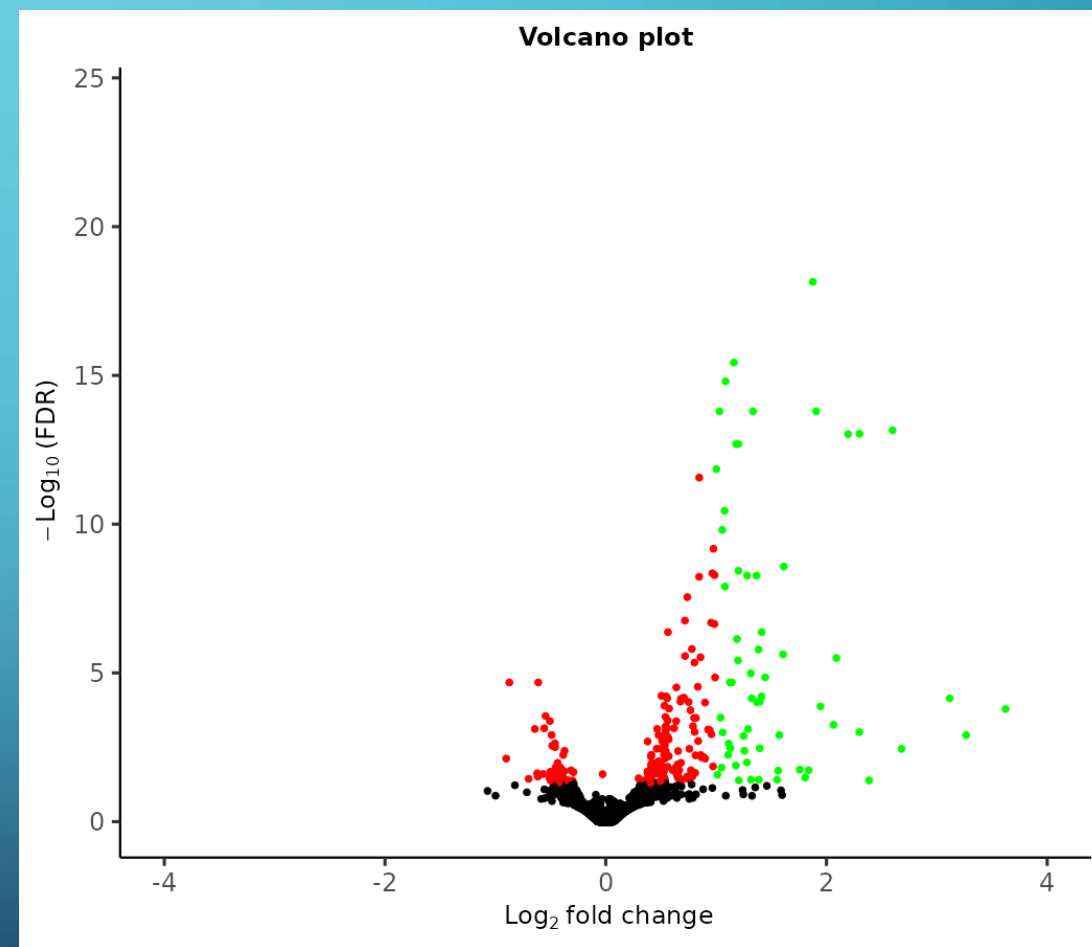
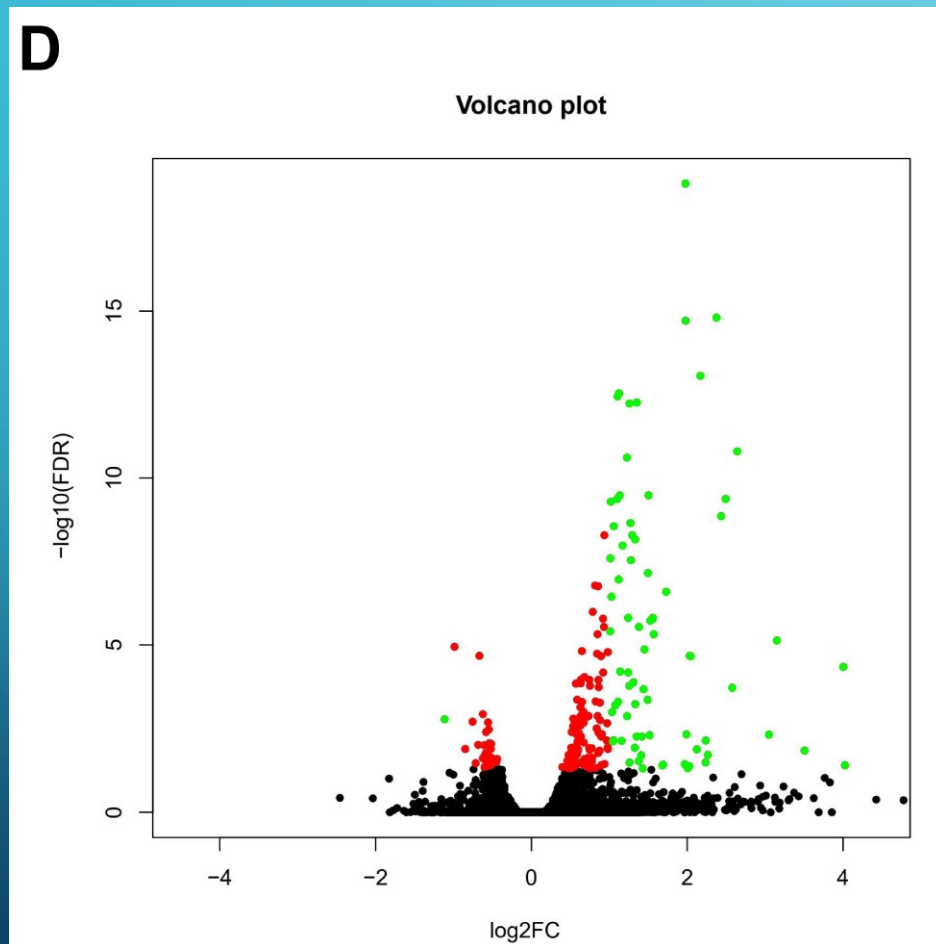


FIG. 1D

We can change some of the formatting, such as grid lines, color scheme, etc., to look like the paper's figure:

```
175 # Enhanced Volcano plot similar to paper's figure
176 EnhancedVolcano(resLFC,
177                 lab = NA,
178                 x = 'log2FoldChange',
179                 y = 'padj',
180                 gridlines.major = FALSE, gridlines.minor = FALSE,
181                 ylab = bquote(~-Log[10] ~ "(FDR)"),
182                 cutoffLineType = "blank",
183                 subtitle=NULL,
184                 colAlpha = 1,
185                 caption = NULL,
186                 col = c("black", "black", "red", "green"),
187                 pCutoff = 5e-02,
188                 xlim=c(-4,4),
189                 legendPosition = 'none') + theme(plot.title=element_text(hjust=0.5))
```

FIG. 1D



RECREATE FIGURES: TABLE 2

Sort by statistical significance (adjusted p-value), add gene symbols, and then save as a CSV file.

```
192 ▾ ##### Exporting a table ##### Table 2 #####
193   # Sort by adjusted p-value
194   resLFC.05fc1 <- arrange(resLFC.05fc1, padj)
195   # Add gene symbols
196   resLFC.05fc1$symbol <- mapIds(org.Hs.eg.db, keys = row.names(resLFC.05fc1),
197                                column = "SYMBOL",
198                                keytype = "ENSEMBL",
199                                multiVals = "first")
200   # Save as CSV
201   write.csv(resLFC.05fc1, file = "DEG_results.csv")
```

TABLE 2

You can open CSV files in a spreadsheet program (e.g. Microsoft Excel) and create a table

Genes	log2FC	p-value	FDR	Function
ATF6	4.87	1.50E-276	2.53E-272	ER stress
SEC24D	1.98	1.80E-23	1.52E-19	Vesicle trafficking
DNAJB9	2.38	2.81E-19	1.58E-15	Heat Shock Protein
HERPUD1	1.98	4.63E-19	1.95E-15	ER-associated degradation
DDIT3	2.17	2.58E-17	8.71E-14	Apoptosis
PDIA4	1.12	1.04E-16	2.92E-13	Protein folding
CREB3L2	1.11	1.49E-16	3.59E-13	Unfolded Protein Response
ARMCX3	1.35	2.60E-16	5.48E-13	Tumor suppression
HSP90B1	1.26	3.14E-16	5.88E-13	Heat Shock Protein
ATF3	2.64	9.49E-15	1.60E-11	Cellular stress response
PGM3	1.23	1.60E-14	2.46E-11	Glycogen synthesis
WIPI1	1.50	2.43E-13	3.31E-10	Protein assembly
MANF	1.13	2.55E-13	3.31E-10	Cell proliferation and death
CHAC1	2.49	3.66E-13	4.23E-10	Unfolded Protein Response
HSPA13	1.10	3.77E-13	4.23E-10	Heat Shock Protein
TMEM263	1.02	4.87E-13	5.13E-10	Unknown
ERO1B	2.44	1.40E-12	1.39E-09	Protein folding
HYOU1	1.27	2.38E-12	2.23E-09	Heat Shock Protein
TMEM50B	1.06	3.14E-12	2.79E-09	Unknown
NUCB2	1.29	6.12E-12	5.16E-09	Calcium homeostasis
PPP1R15A	1.33	9.02E-12	6.91E-09	Cellular stress response
EIF2AK3	1.17	1.45E-11	1.06E-08	Unfolded Protein Response
SELENOK	1.01	3.64E-11	2.56E-08	The Selenoprotein gene
ARFGAP3	1.28	4.31E-11	2.90E-08	GTPase-activating protein
GABARAPL1	1.50	1.09E-10	7.05E-08	Autophagosome assembly
PNRC1	1.12	1.76E-10	1.10E-07	Unknown
HSPA5	1.73	4.59E-10	2.58E-07	Heat Shock Protein
SEL1L	1.03	6.66E-10	3.62E-07	Protein degradation
ADM2	1.56	3.13E-09	1.55E-06	Peptide hormone
SESN2	1.24	3.11E-09	1.55E-06	Cell growth and survival
FBXO16	1.56	3.49E-09	1.64E-06	Protein degradation
GADD45A	1.53	4.12E-09	1.87E-06	Cellular stress response
FKBP14	1.38	6.70E-09	2.90E-06	Protein folding
SLC33A1	1.01	9.31E-09	3.92E-06	Acetylation-mediated secretion
TSPYL2	1.57	1.20E-08	4.81E-06	Chromatin remodeling
CDH15	3.15	1.88E-08	7.37E-06	Calcium-dependent cell adhesion
DNAJC3	1.45	3.66E-08	1.37E-05	Heat Shock Protein
TRPV6	2.03	6.31E-08	2.13E-05	Calcium channel
ZSCAN12P1	2.04	6.73E-08	2.18E-05	Pseudogene

J7	A	B	C	D	E	F	G
1		baseMean	log2FoldC	lfcSE	pvalue	padj	symbol
2	ENSG0000	17939.39	5.226118	0.13129	0	0	ATF6
3	ENSG0000	1221.688	1.876807	0.197257	9.39E-23	7.18E-19	SEC24D
4	ENSG0000	768.5579	1.161128	0.131987	7.26E-20	3.70E-16	PGM3
5	ENSG0000	12808.49	1.085462	0.126206	4.15E-19	1.58E-15	PDIA4
6	ENSG0000	1797.897	1.907256	0.229903	5.27E-18	1.61E-14	HERPUD1
7	ENSG0000	916.8799	1.334793	0.161336	6.49E-18	1.61E-14	ARMCX3
8	ENSG0000	1413.192	1.030051	0.124686	7.39E-18	1.61E-14	TMEM263
9	ENSG0000	217.8227	2.599182	0.320688	3.69E-17	7.05E-14	ATF3
10	ENSG0000	1695.389	2.300049	0.287102	5.37E-17	9.13E-14	DNAJB9
11	ENSG0000	455.1853	2.196027	0.274508	6.17E-17	9.44E-14	DDIT3
12	ENSG0000	722.5884	1.18127	0.149599	1.46E-16	2.03E-13	PNRC1
13	ENSG0000	21907.09	1.202364	0.152447	1.59E-16	2.03E-13	HSP90B1
14	ENSG0000	1147.738	1.002633	0.131444	1.20E-15	1.42E-12	CREB3L2
15	ENSG0000	1538.496	1.076477	0.14992	3.46E-14	3.53E-11	MANF
16	ENSG0000	1174.812	1.055681	0.151657	1.63E-13	1.56E-10	HSPA13
17	ENSG0000	150.5654	1.613806	0.247013	3.15E-12	2.67E-09	TVP23C-CDRT4
18	ENSG0000	8262.614	1.202108	0.185686	4.59E-12	3.70E-09	HYOU1
19	ENSG0000	363.4676	1.365552	0.213692	8.03E-12	5.35E-09	WIPI1
20	ENSG0000	587.4781	1.281187	0.200785	8.40E-12	5.35E-09	PPP1R15A
21	ENSG0000	26.04056	9.637883	2.841626	8.34E-12	5.35E-09	GH1
22	ENSG0000	1427.052	1.080048	0.173034	2.13E-11	1.25E-08	EIF2AK3
23	ENSG0000	128.9644	1.413685	0.251607	8.73E-10	4.25E-07	GABARAPL1

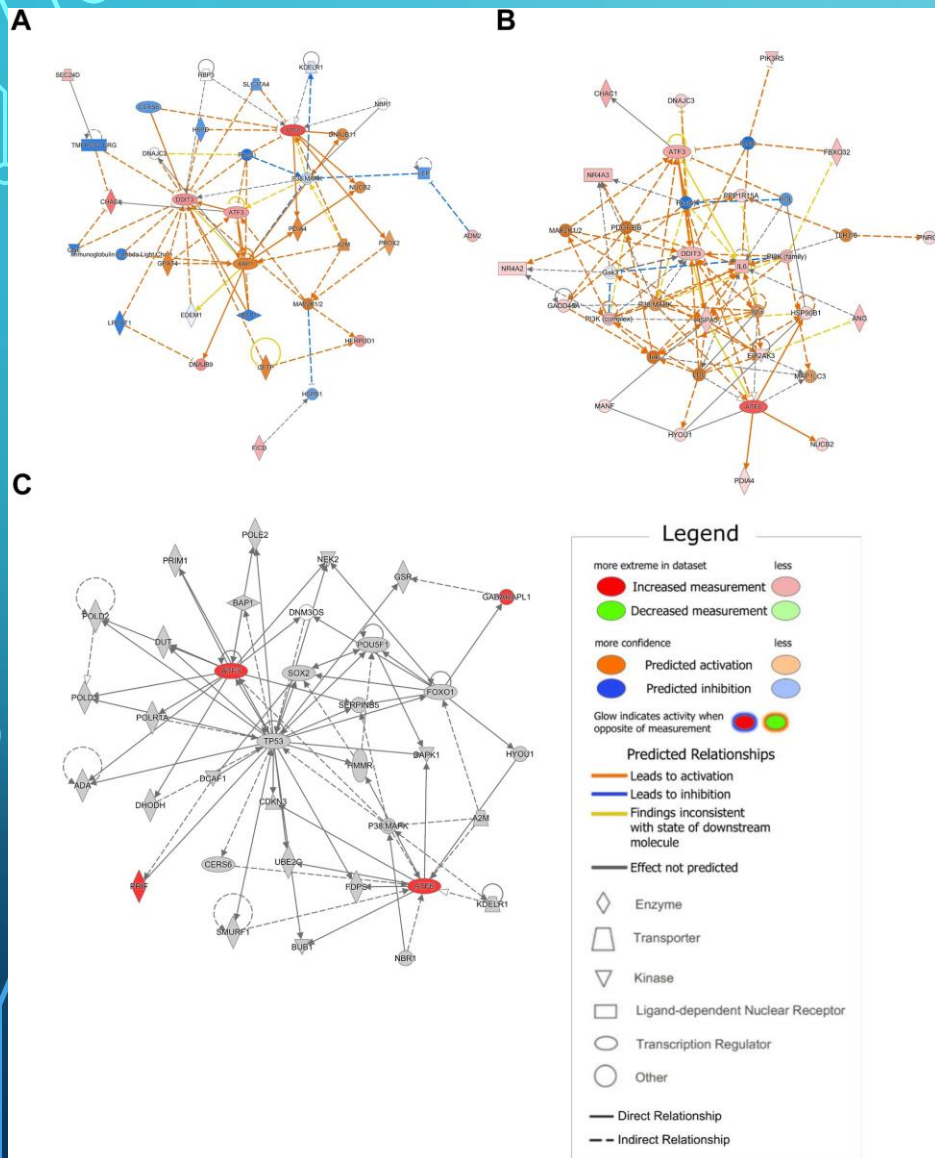
RECREATE FIGURES: FIG. 2



Use the STRINGdb package to see PPI networks of the highly differentially expressed genes

```
204 ##### Protein-protein interaction (PPI) Networks using STRING ##### Fig. 2 #####
205 # load the STRING database package
206 library(STRINGdb)
207 # Fetch the Homo sapiens database (#9606)
208 string_db <- STRINGdb$new(species=9606)
209 # map the gene names to the STRING database identifiers using the "map" method
210 mapped <- string_db$map(vsd_df, "symbol", removeUnmappedRows = TRUE )
211 # Extract mapped genes
212 hits <- mapped$STRING_id
213 # Plot network
214 string_db$plot_network(hits)
```

FIG. 2



proteins: 65
interactions: 124
expected interactions: 12 (p-value: 0)

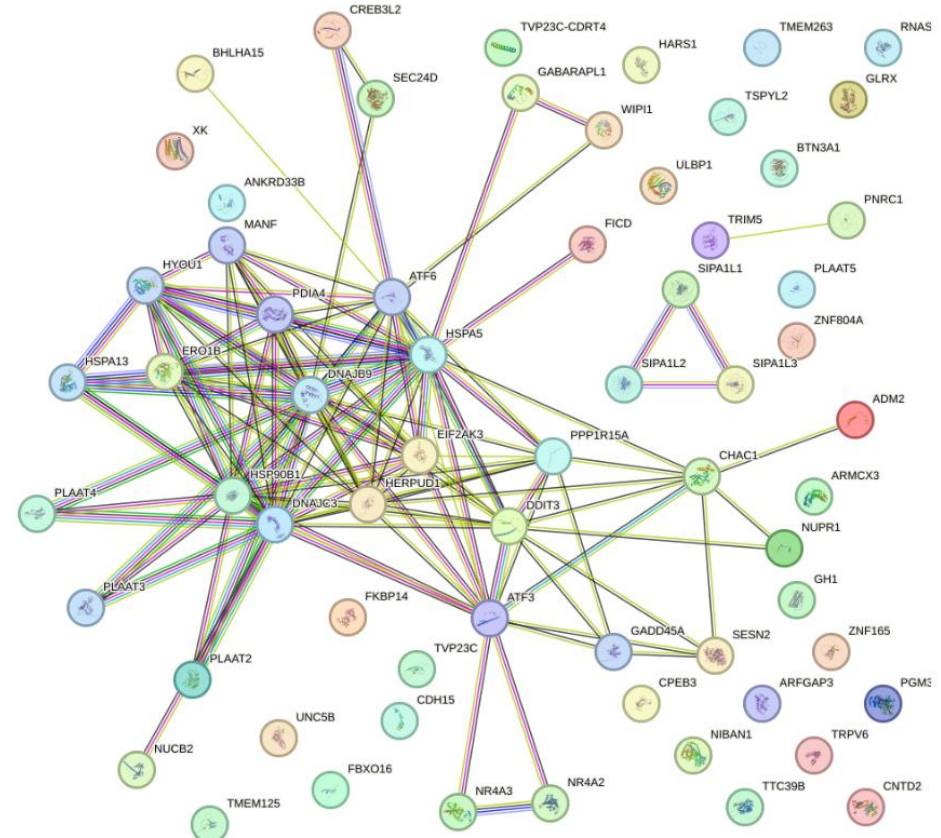


FIG. 2

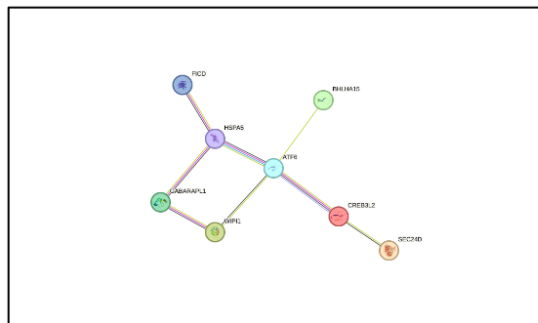


We can cluster the results:

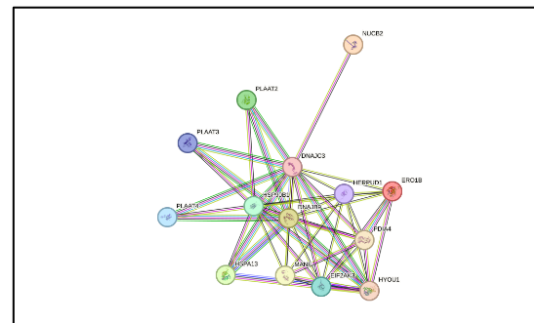
```
216 # cluster interactions and plot first 4 clusters
217 clustersList<- string_db$get_clusters(mapped$STRING_id)
218 par(mfrow=c(2,2))
219 for(i in seq(1:4)){
220     string_db$plot_network(clustersList[[i]])
221 }
222 #reset plotting device
223 dev.off()
```

FIG. 2

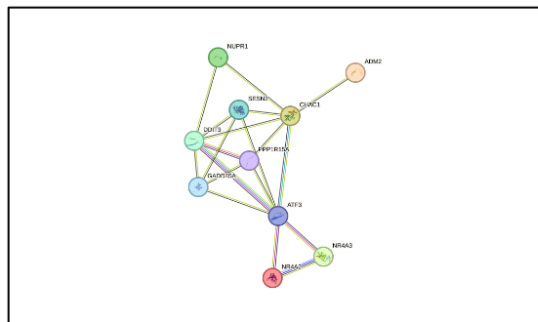
proteins: 8
interactions: 8
expected interactions: 1 (p-value: 7.89e-08)



proteins: 14
interactions: 49
expected interactions: 1 (p-value: 0)



proteins: 10
interactions: 19
expected interactions: 0 (p-value: 0)



proteins: 3
interactions: 3
expected interactions: 0 (p-value: 6.35e-10)

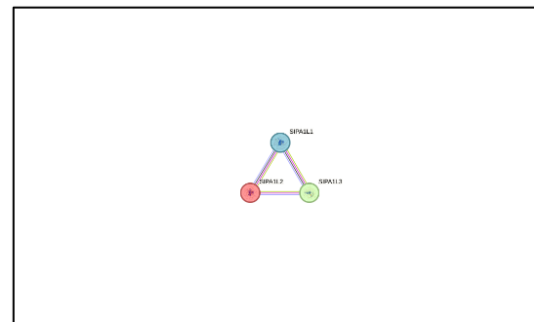


FIG. 2



We can see all known interacting proteins with a particular protein/gene

```
225 # Get a list of proteins known to interact with a protein (or proteins) of interest
226 atf6 = string_db$mp("atf6")
227 string_db$get_neighbors(atf6)
```

```
[1] "9606.ENSP00000078445" "9606.ENSP000000158771" "9606.ENSP000000185150" "9606.ENSP000000199389"
[5] "9606.ENSP000000200453" "9606.ENSP000000212015" "9606.ENSP000000216037" "9606.ENSP000000216038"
[9] "9606.ENSP000000216117" "9606.ENSP000000222254" "9606.ENSP000000225577" "9606.ENSP000000226574"
[13] "9606.ENSP000000226760" "9606.ENSP000000229795" "9606.ENSP000000233057" "9606.ENSP000000243253"
[17] "9606.ENSP000000247461" "9606.ENSP000000247668" "9606.ENSP000000249356" "9606.ENSP000000251047"
[21] "9606.ENSP000000251337" "9606.ENSP000000251968" "9606.ENSP000000252486" "9606.ENSP000000252804"
[25] "9606.ENSP000000254322" "9606.ENSP000000254325" "9606.ENSP000000255040" "9606.ENSP000000256383"
[29] "9606.ENSP000000256497" "9606.ENSP000000256797" "9606.ENSP000000259253" "9606.ENSP000000259512"
[33] "9606.ENSP000000261735" "9606.ENSP000000261942" "9606.ENSP000000262039" "9606.ENSP000000262139"
[37] "9606.ENSP000000262238" "9606.ENSP000000262455" "9606.ENSP000000263025" "9606.ENSP000000263341"
```

STRING WEBSERVER: STRING-DB.ORG



Version: 12.0

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STRING

[Search](#) | [Download](#) | [Help](#) | [My Data](#)

Protein by name >

Multiple proteins >

Proteins by sequences >

Proteins with Values/Ranks >

Protein families ("COGs") >

Pathway / Process / Disease New >

Add organism New >

Organisms >

Examples >

Random entry >

SEARCH

Single Protein by Name / Identifier

Protein Name: (examples: [#1](#) [#2](#) [#3](#))

Organisms:

auto-detect ▼

[Advanced Settings](#)

SEARCH

NETWORK ANALYSIS: CYTOSCAPE

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Analyze Your Genes With NDEx
iQuery

Type or paste your list of genes here...



RECREATE FIGURES: PATHWAY ANALYSIS

Here we'll use the 'clusterProfiler' package to perform Gene Ontology (GO) analysis

```
230 ▾ ##### Functional Enrichment with clusterProfiler ##### Table 4 #####  
231 # Load package  
232 library(clusterProfiler)  
233 # Create gene list from previous data frame (Ensembl IDs)  
234 genes <- row.names(resLFC.05fc1)  
235 ## Gene Ontology (GO) over-representation analysis (ORA)  
236 ## Biological Process (BP) ## (also CC and MF available)  
237 go_bp_over <- enrichGO(gene = genes,  
238                        keyType = 'ENSEMBL',  
239                        OrgDb = org.Hs.eg.db,  
240                        universe = row.names(res),  
241                        ont = "BP",  
242                        readable = TRUE)  
243 # Check results  
244 head(go_bp_over)
```

PATHWAY ANALYSIS: CLUSTERPROFILER



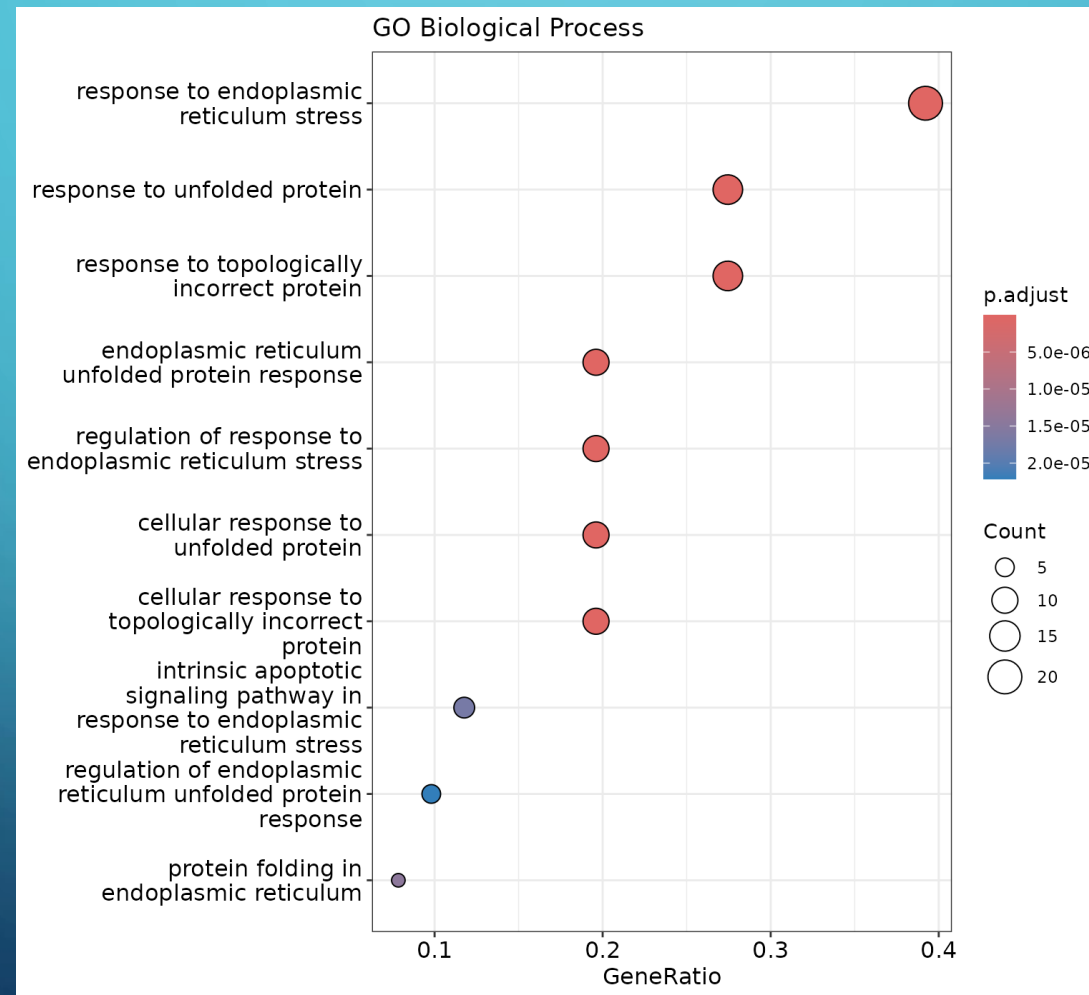
ID		Description	GeneRatio	BgRatio		
GO:0034976	GO:0034976	response to endoplasmic reticulum stress	20/51	244/11759		
GO:0006986	GO:0006986	response to unfolded protein	14/51	125/11759		
GO:0035966	GO:0035966	response to topologically incorrect protein	14/51	142/11759		
GO:0030968	GO:0030968	endoplasmic reticulum unfolded protein response	10/51	77/11759		
GO:1905897	GO:1905897	regulation of response to endoplasmic reticulum stress	10/51	78/11759		
GO:0034620	GO:0034620	cellular response to unfolded protein	10/51	88/11759		
	RichFactor	FoldEnrichment	zScore	pvalue	p.adjust	qvalue
GO:0034976	0.08196721	18.89907	18.64678	4.388148e-21	4.967383e-18	4.471292e-18
GO:0006986	0.11200000	25.82369	18.41481	1.036910e-16	5.868911e-14	5.282784e-14
GO:0035966	0.09859155	22.73212	17.19530	6.452830e-16	2.434868e-13	2.191698e-13

PATHWAY ANALYSIS: CLUSTERPROFILER



The 'Dot Plot' is a popular way of visualizing enrichment results:

```
246 # Visualize enriched GO terms with a Dot Plot
247 dotplot(go_bp_over) + ggtitle("GO Biological Process")
```



PATHWAY ANALYSIS: ENRICHR

<https://maayanlab.cloud/Enrichr>



Enrichr

[Login](#) | [Register](#)

88,691,339 sets analyzed

524,700 terms

236 libraries

Analyze

[What's new?](#)

[Libraries](#)

[Gene search](#)

[Term search](#)

[About](#)

[Help](#)

Input data

Expand a gene, a term, or a variant into a gene set:

e.g. STAT3, breast cancer, or rs28897756



Try an example

Include the top 100 most relevant genes



Drop a file or paste a set of Entrez gene symbols on each row in the textbox below. You can try a gene set **example**. Also, you can now try adding a **background**.

Drop a file or paste a set of valid Entrez gene symbols (e.g. STAT3) on each row in the text-box

0 gene(s) entered

PATHWAY ANALYSIS: ENRICHR



Enrichr

[Login](#) | [Register](#)

88,853,869 sets analyzed

524,700 terms

236 libraries

[Analyze](#)[What's new?](#)[Libraries](#)[Gene search](#)[Term search](#)[About](#)[Help](#)

Gene-set Library	Terms	Gene Coverage	Genes per Term	
MAGMA_Drugs_and_Diseases	1395	2183	18	↓
MAGNET_2023	72	12853	485	↓
MCF7_Perturbations_from_GEO_down	401	15022	290	↓
MCF7_Perturbations_from_GEO_up	401	15676	310	↓
Metabolomics_Workbench_Metabolites_2022	233	936	15	↓

PATHWAY ANALYSIS: ENRICHR



```
250 ▾ ##### Functional Enrichment with EnrichR #####
251 # Load package
252 library(enrichR)
253 # Fetch the MCF7_Perturbations_from_GEO_up database
254 db_mcf7 <- c("MCF7_Perturbations_from_GEO_up")
255 # Run analysis
256 enriched <- enrichr(vsd_df$symbol, db_mcf7, background = res$symbol)
257 # Plot results
258 plotEnrich(enriched[["MCF7_Perturbations_from_GEO_up"]], showTerms = 20, numChar = 40,
259             y = "Count", orderBy = "Adjusted.P.value", title = "MCF7 Perturbations from GEO Up")
```

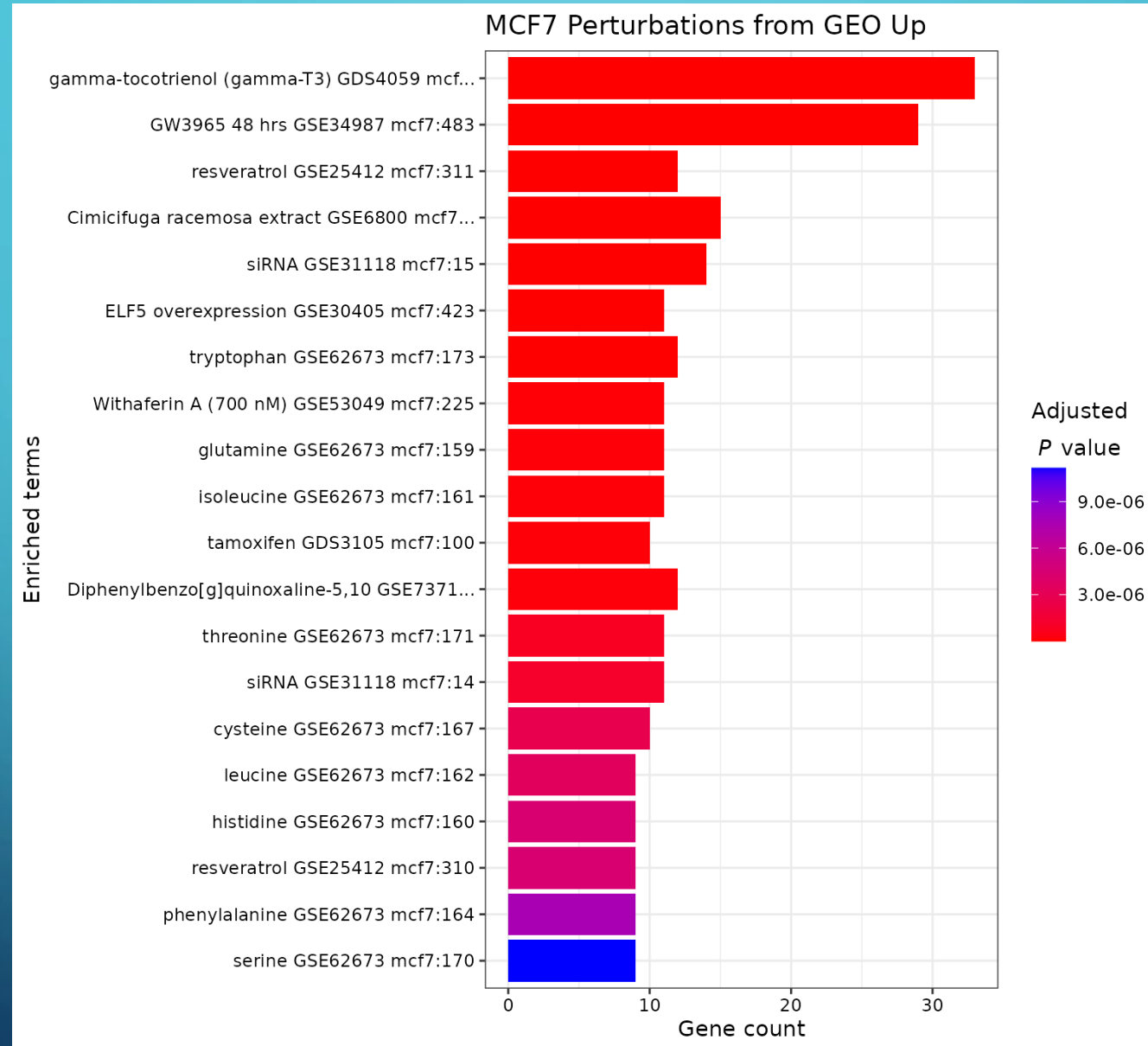
PATHWAY ANALYSIS: ENRICHR



Here, we'll use a relevant database to perform an enrichment analysis (MCF7 cells were used in the publication) and plot the results:

```
250 ▾ ##### Functional Enrichment with EnrichR #####
251 # Load package
252 library(enrichR)
253 # Fetch the MCF7_Perturbations_from_GEO_up database
254 db_mcf7 <- c("MCF7_Perturbations_from_GEO_up")
255 # Run analysis
256 enriched <- enrichr(vsd_df$symbol, db_mcf7, background = res$symbol)
257 # Plot results
258 plotEnrich(enriched[["MCF7_Perturbations_from_GEO_up"]], showTerms = 20, numChar = 40,
259            y = "Count", orderBy = "Adjusted.P.value", title = "MCF7 Perturbations from GEO Up")
```

PATHWAY ANALYSIS: ENRICH



RECREATE FIGURES: FIG. 4

Here, we'll use an R package called 'fgsea' to run Gene Set Enrichment Analysis (GSEA). First, we need to load the necessary packages and create an input file:

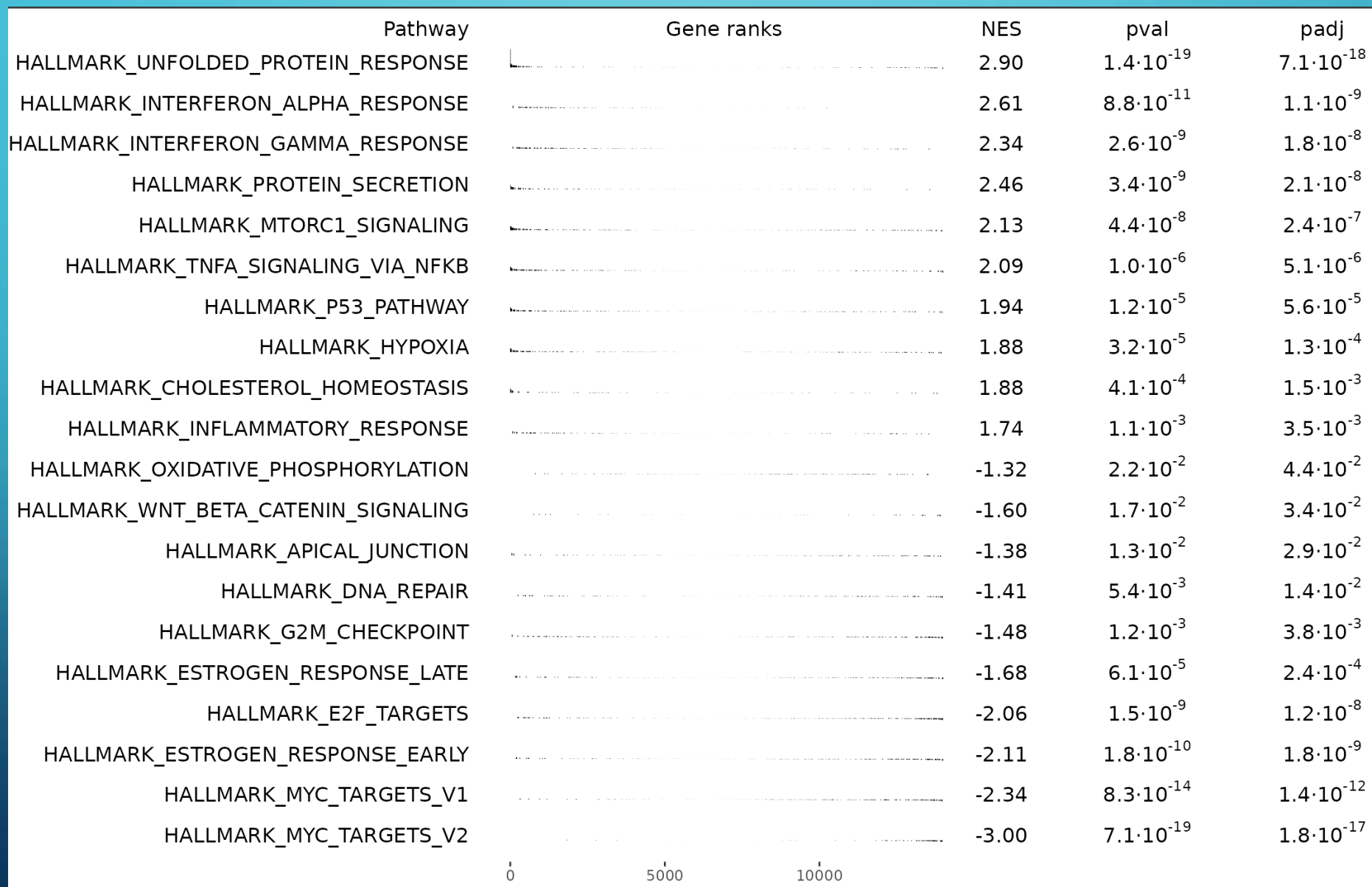
```
262 ▾ ##### fgsea package #####  
263 library(fgsea)  
264 library(msigdb)  
265 # Create fgsea input  
266 res_filtered <- as.data.frame(res) %>% dplyr::select(symbol,stat) %>% na.omit() %>% distinct %>% group_by(symbol)  
267 fgsea_input <- res_filtered$stat  
268 names(fgsea_input) <- res_filtered$symbol
```

GENE SET ENRICHMENT ANALYSIS: FGSEA

Then, we can run the analysis and plot the results:

```
269 # Run fgsea
270 hallmarks = msigdbr(species = "human", collection = "H")
271 hallmarks_list = split(hallmarks$gene_symbol, f = hallmarks$gs_name)
272 fgsea_res <- fgsea(pathways=hallmarks_list, fgsea_input, minSize = 5, maxSize = 500)
273 fgsea_res_ordered <- fgsea_res[order(padj), ]
274 # The below line will save the results as a .csv file (uncomment to run)
275 #fwrite(fgsea_res_ordered, file = "fgsea_table_GSE231409.csv", sep = ",", sep2 = c("", " ", ""))
276 topPathwaysUp <- fgsea_res[ES > 0][head(order(pval), n=10), pathway]
277 topPathwaysDown <- fgsea_res[ES < 0][head(order(pval), n=10), pathway]
278 topPathways <- c(topPathwaysUp, rev(topPathwaysDown))
279 # The below line will save the plot as a .tiff file (uncomment to run)
280 #tiff("fgsea_hallmarks.tiff", width = 800, height = 800)
281 plotGseaTable(hallmarks_list[topPathways], fgsea_input, fgsea_res)
```

GENE SET ENRICHMENT ANALYSIS: FGSEA



RECREATE FIGURES: FIG. 4

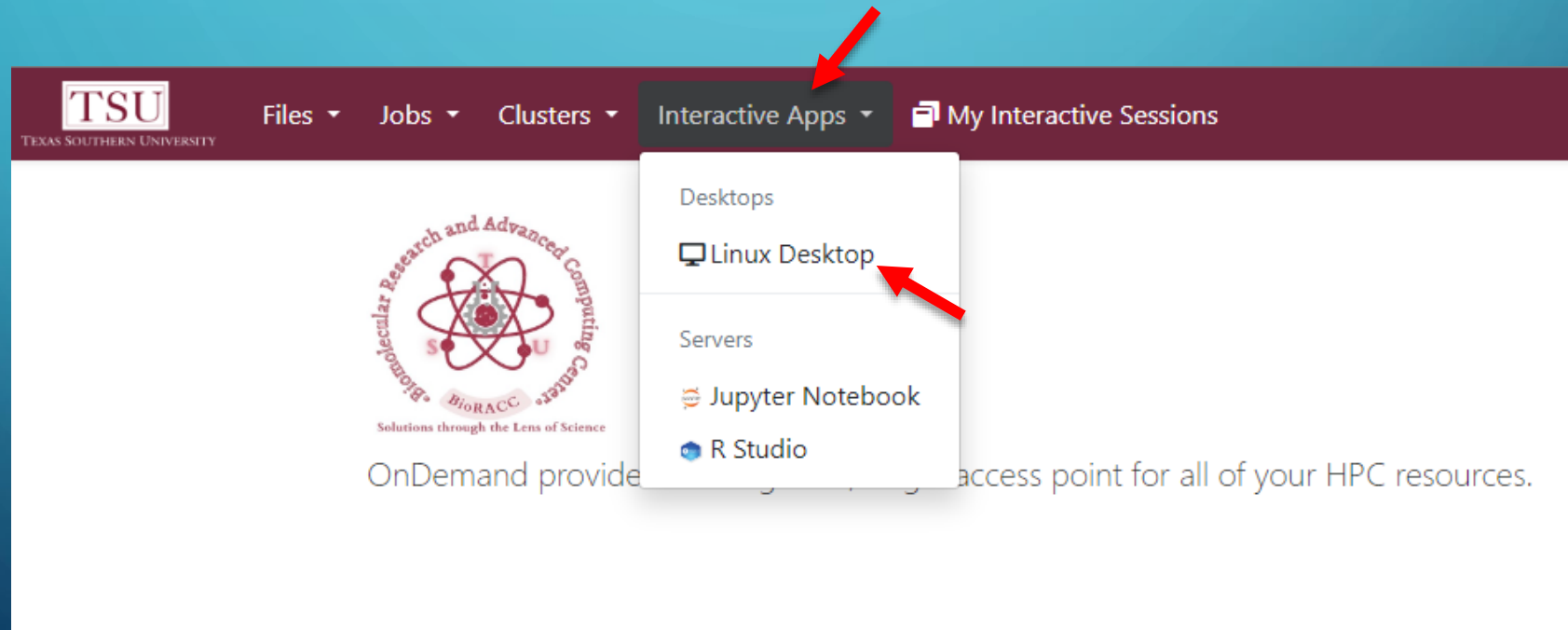
Here we'll format our DE results to use for GSEA. The input cannot have any duplicate values. We'll use the test statistic that comes from the DE analysis to rank the genes:

```
284 ##### GSEA (BROAD) ##### Fig. 4 #####
285 # Process results to use with GSEA
286 gsea_input <- as.data.frame(res) %>%
287   rownames_to_column(var = "gene_id") %>%
288   dplyr::select(gene_id,stat) %>%
289   na.omit() %>%
290   distinct %>%
291   group_by(gene_id) %>%
292   summarise(stat=mean(stat)) %>%
293   filter(!duplicated(stat))
294 #save file
295 write.table(gsea_input, file = "gsea_input.rnk", quote = F, sep = "\t", row.names=F, col.names = F)
```

RECREATE FIGURES: FIG. 4

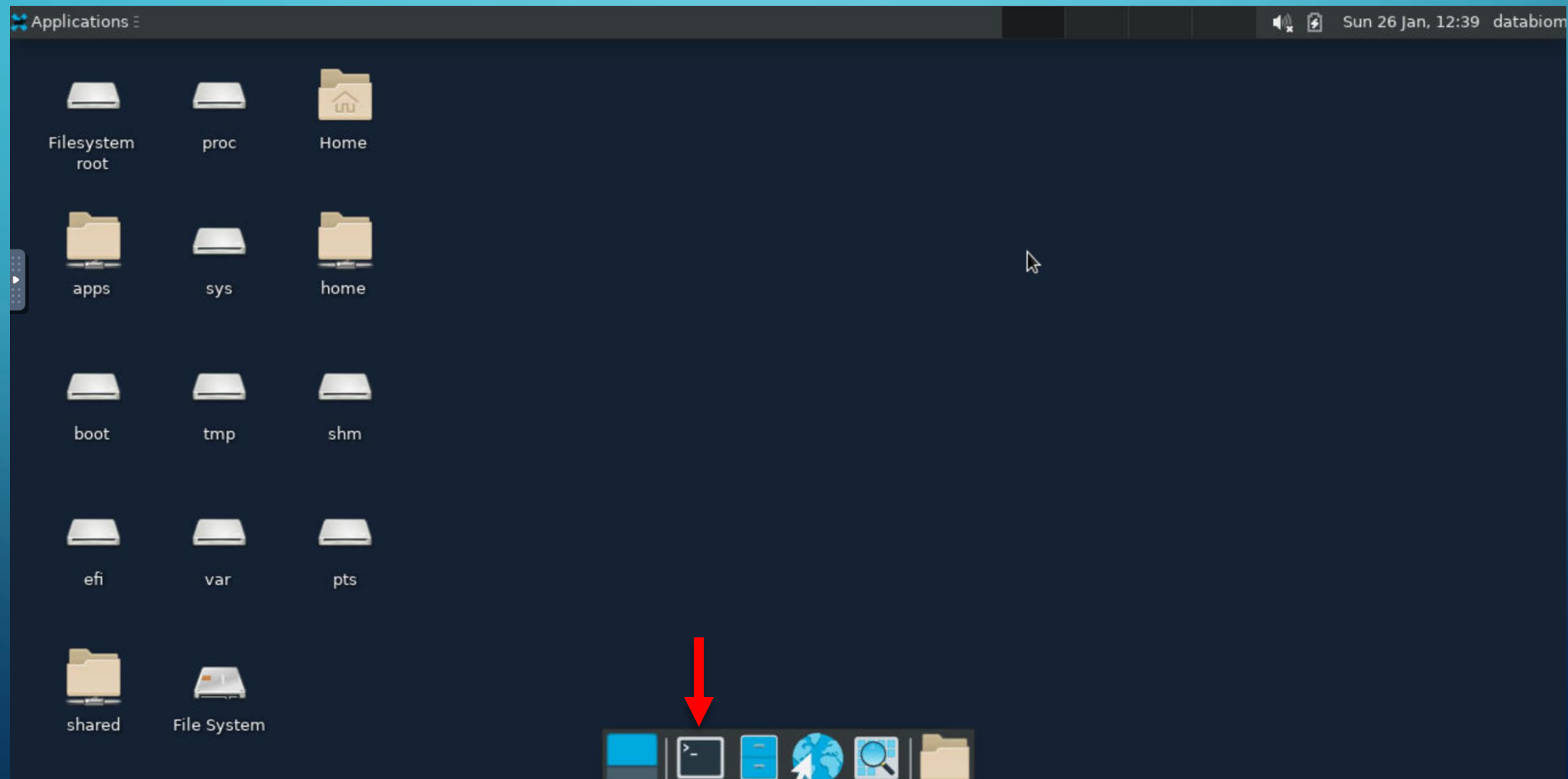
Now that we have our input file for GSEA, we need to switch over to a Linux Desktop session to access the Broad Institute GSEA program.

```
297 ▾ ##### Request a Linux Desktop session to run Broad GSEA GUI Application #####  
298 ▾ ##### see PowerPoint for details #####
```



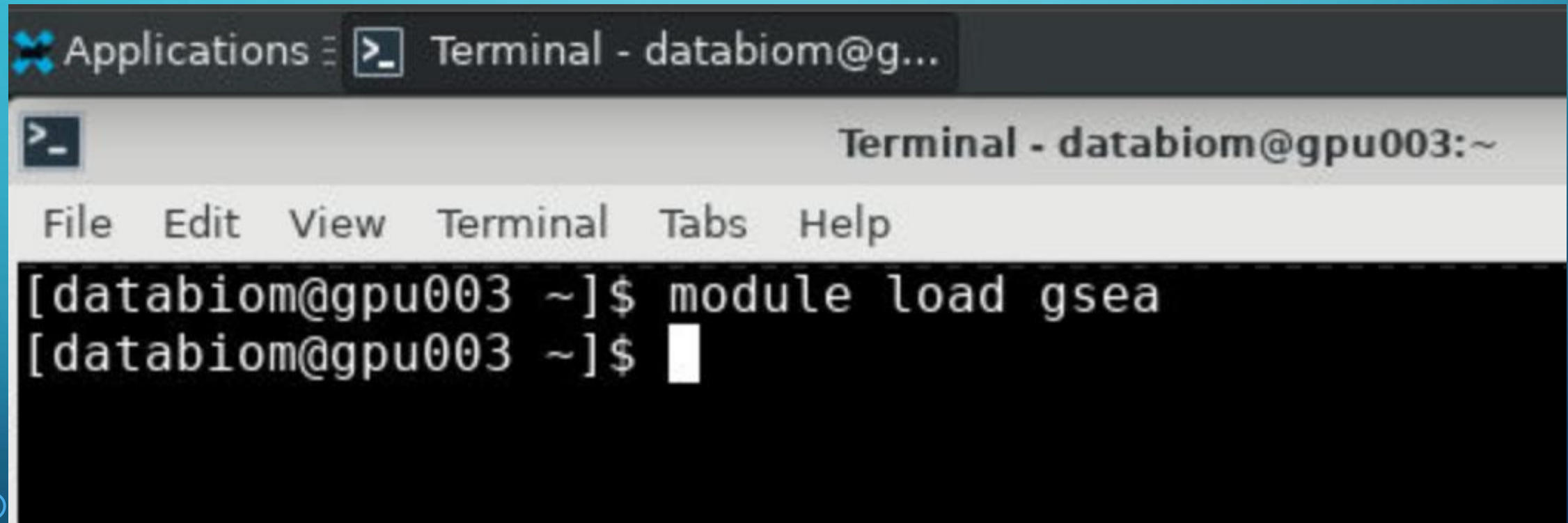
RECREATE FIGURES: FIG. 4

Open a terminal



RECREATE FIGURES: FIG. 4

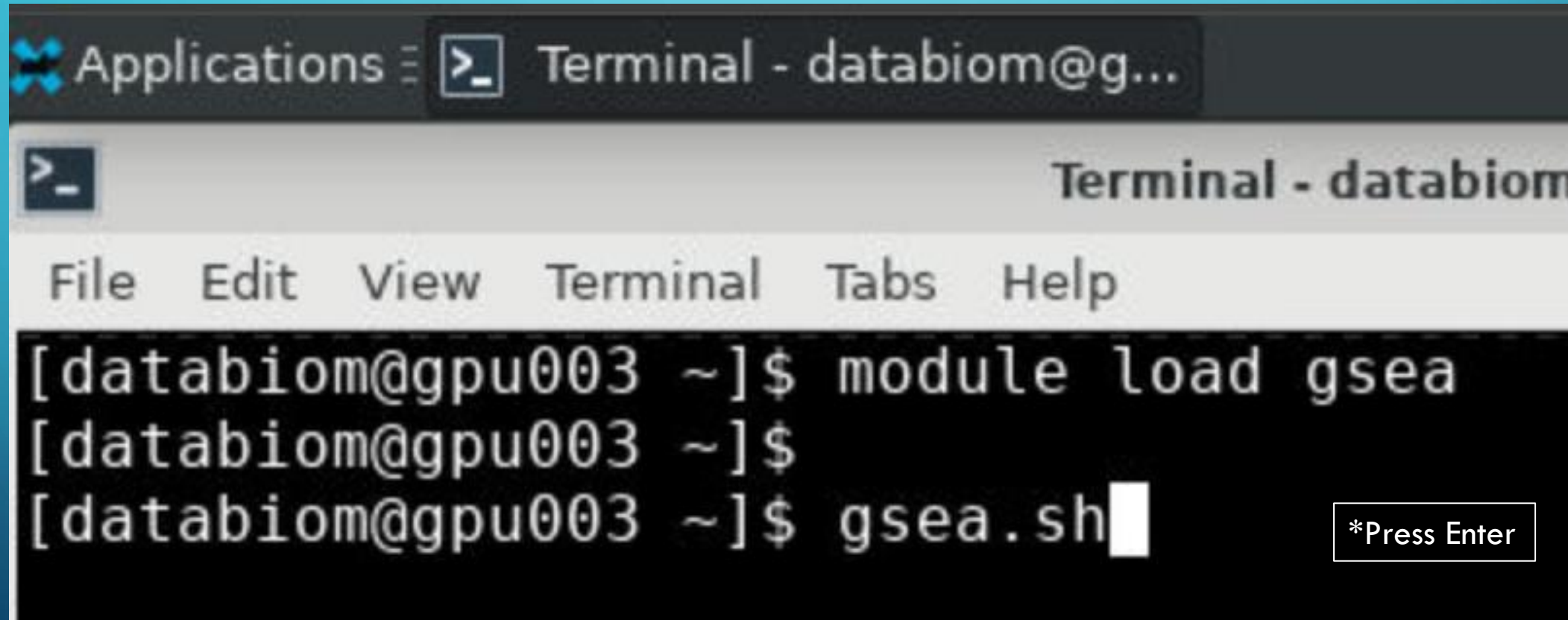
Load the gsea module:



```
Applications ▢ Terminal - databiom@g...  
Terminal - databiom@gpu003:~  
File Edit View Terminal Tabs Help  
[databiom@gpu003 ~]$ module load gsea  
[databiom@gpu003 ~]$
```

RECREATE FIGURES: FIG. 4

Run the gsea.sh script that will open the graphical user interface (GUI):



```
Applications ▢ Terminal - databiom@g...  
Terminal - databiom  
File Edit View Terminal Tabs Help  
[databiom@gpu003 ~]$ module load gsea  
[databiom@gpu003 ~]$  
[databiom@gpu003 ~]$ gsea.sh
```

*Press Enter

RECREATE FIGURES: FIG. 4

GSEA GUI:



Applications GSEA 4.3.2 (Gene set e... Terminal - databiom@g... Sat 15 Mar, 14:18 databiom

GSEA 4.3.2 (Gene set enrichment analysis)

File Downloads Help

Steps in GSEA analysis

- Load data
- Run GSEA
- Leading edge analysis
- Enrichment Map Visualization

Tools

- Run GSEAPreranked
- Collapse Dataset
- Chip2Chip mapping
- Analysis history

GSEA reports

Processes: click 'status' field for results

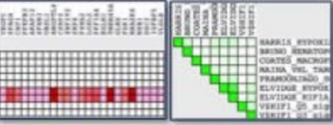
Show results folder

2:18:40 PM

48M of 2048M

Home

Steps in GSEA

- 1. What you need for GSEA**
 - Expression data set
 - Phenotype annotation
 - Gene sets – use MSigDB or your own gene sets
- 2. Run GSEA**
 - Start with default parameters
 - If you want to collapse probes to genes, specify chip platform
- 3. View results**
- 4. Leading edge analysis**
 - Leading edge finds genes driving enrichment results

Gene Set Tools

Chip2Chip mapping

- Convert gene sets between platforms

 Chip2Chip mapping

Explore MSigDB gene sets

- See the online tools and data at www.msigdb.org
- Search the database of thousands of gene sets
- Browse the gene sets by name
- Find overlapping gene sets
- Export gene sets

Getting Help

GSEA web site:

www.gsea-msigdb.org

Contact the GSEA team:

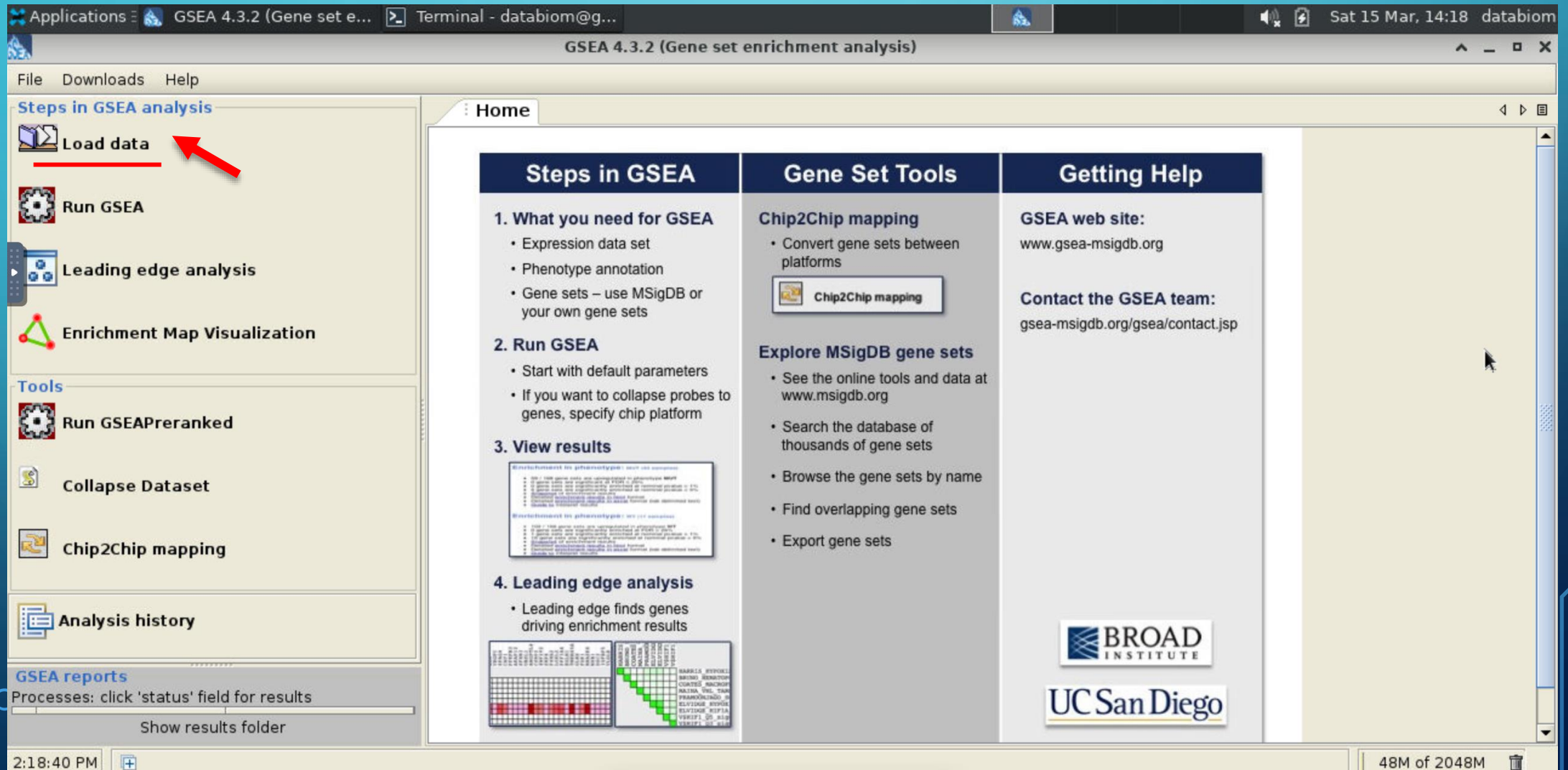
gsea-msigdb.org/gsea/contact.jsp

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RECREATE FIGURES: FIG. 4

Click “**Load data**” from the menu on the left:



Applications GSEA 4.3.2 (Gene set e... Terminal - databiom@g... Sat 15 Mar, 14:18 databiom

GSEA 4.3.2 (Gene set enrichment analysis)

File Downloads Help

Steps in GSEA analysis

 **Load data** 

 **Run GSEA**

 **Leading edge analysis**

 **Enrichment Map Visualization**

Tools

 **Run GSEAPreranked**

 **Collapse Dataset**

 **Chip2Chip mapping**

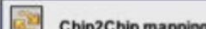
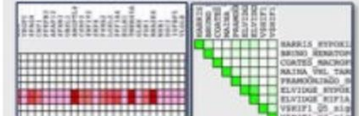
 **Analysis history**

GSEA reports

Processes: click 'status' field for results

Show results folder

Home

Steps in GSEA	Gene Set Tools	Getting Help
1. What you need for GSEA - Expression data set - Phenotype annotation - Gene sets – use MSigDB or your own gene sets	Chip2Chip mapping Convert gene sets between platforms 	GSEA web site: www.gsea-msigdb.org
2. Run GSEA - Start with default parameters - If you want to collapse probes to genes, specify chip platform	Explore MSigDB gene sets - See the online tools and data at www.msigdb.org - Search the database of thousands of gene sets - Browse the gene sets by name - Find overlapping gene sets - Export gene sets	Contact the GSEA team: gsea-msigdb.org/gsea/contact.jsp
3. View results 		
4. Leading edge analysis Leading edge finds genes driving enrichment results 		

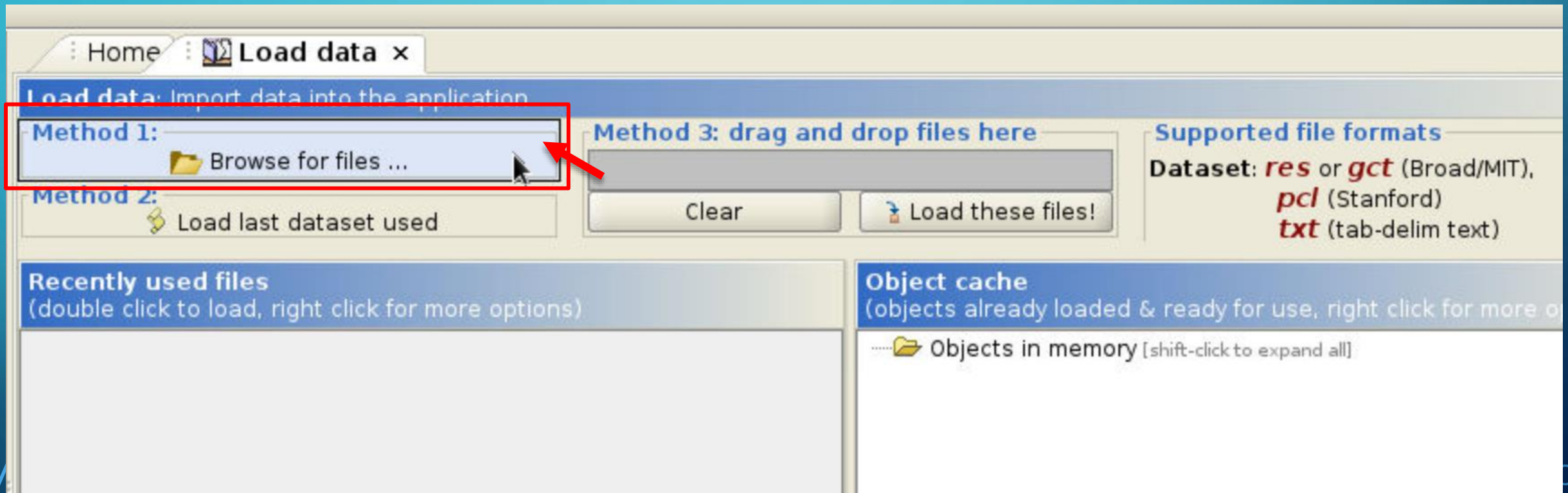
BROAD INSTITUTE

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2:18:40 PM 48M of 2048M

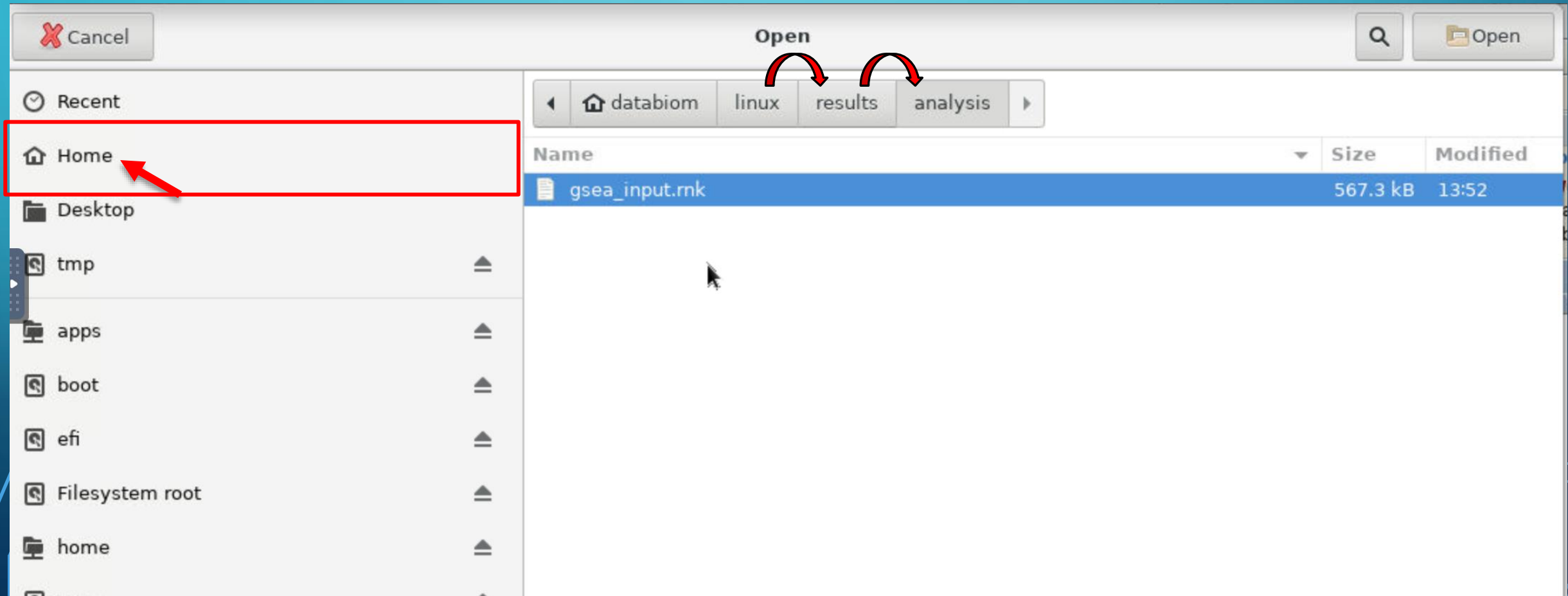
RECREATE FIGURES: FIG. 4

Click on the “Browse for files ...” button



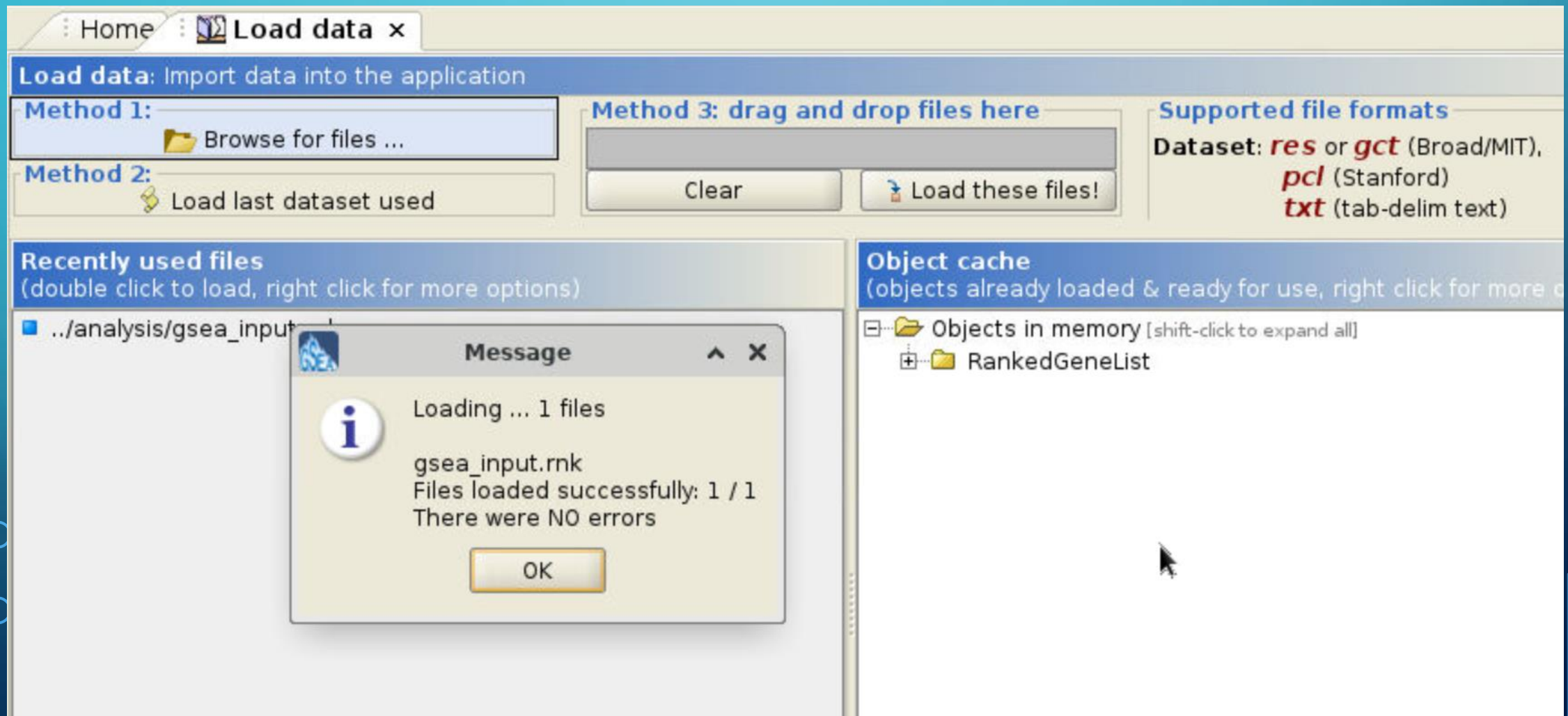
RECREATE FIGURES: FIG. 4

On the left side, click on “Home”. Then on the right side, navigate to the gsea_input.rnk file. It should be in the path linux/results/analysis/gsea_input.rnk



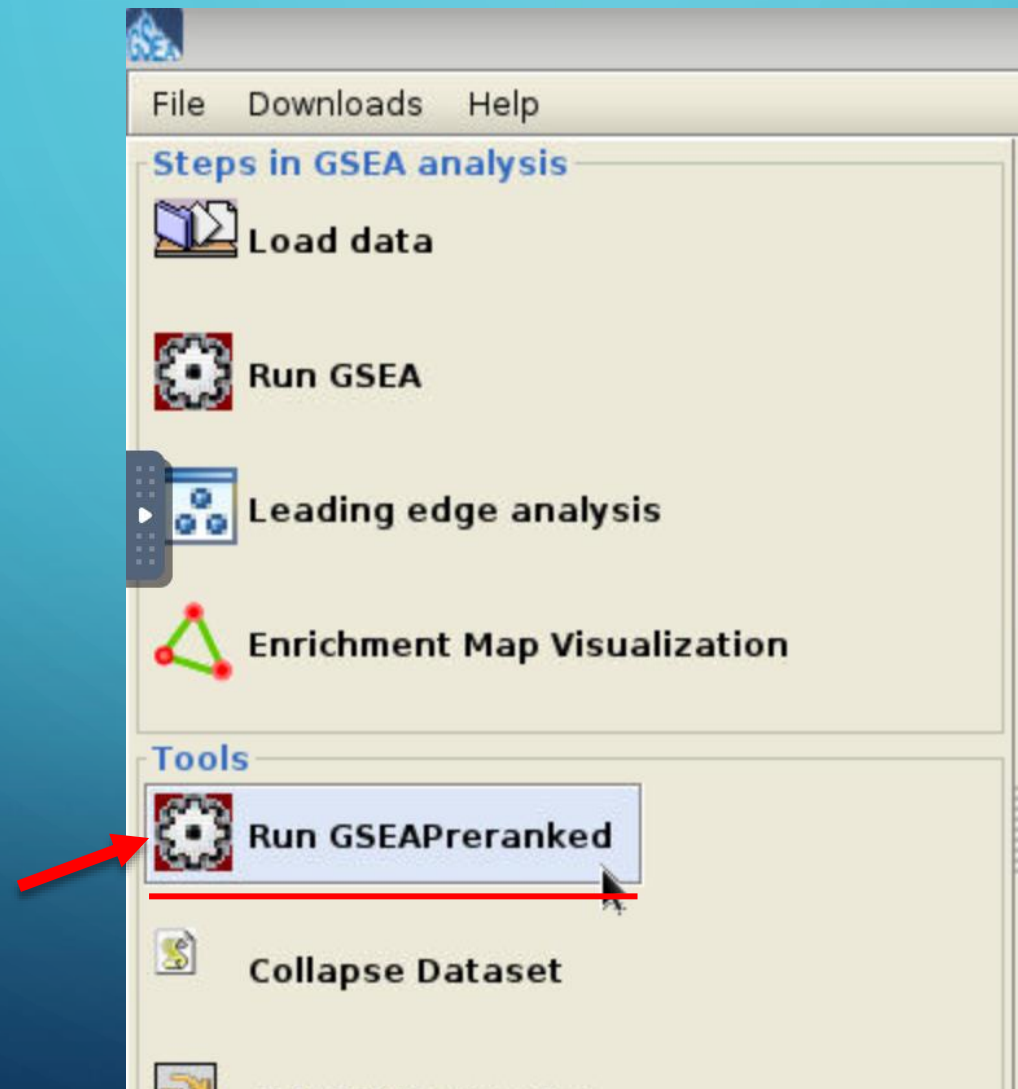
RECREATE FIGURES: FIG. 4

You should get the following pop-up message (click OK):



RECREATE FIGURES: FIG. 4

Click the “**Run GSEAPreranked**” in the menu on the left:



RECREATE FIGURES: FIG. 4

Click on the button with three dots (...) for the “**Gene sets database**” dropdown:

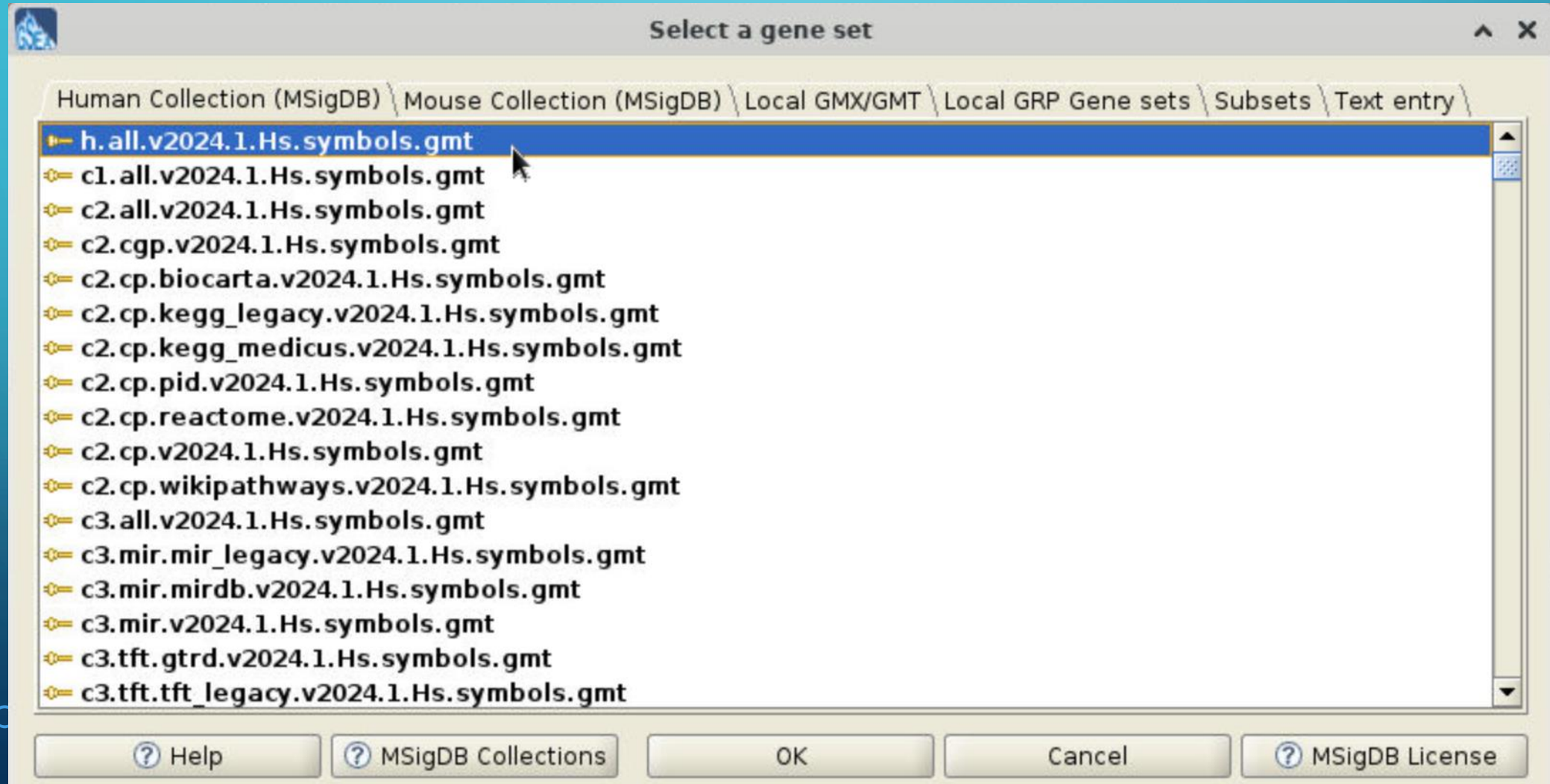


The screenshot shows the GSEA web interface with the following fields and controls:

- Required fields:**
 - Gene sets database:** A text input field with a three-dot button (highlighted by a red box and a red arrow) to the right.
 - Number of permutations:** A dropdown menu set to 1000.
 - Ranked List:** A dropdown menu set to gsea_input [16584 names].
 - Collapse/Remap to gene symbols:** A dropdown menu set to Remap_Only.
 - Chip platform:** A text input field with a three-dot button to the right.
- Basic fields:** A section with a "Show" button.
- Advanced fields:** A section with a "Show" button.

RECREATE FIGURES: FIG. 4

Select (double-click) the **first set**, “**h.all.v2024.1.Hs.symbols.gmt**”



RECREATE FIGURES: FIG. 4

Click on the drop-down arrow for the “**Collapse/Remap to gene symbols**”:

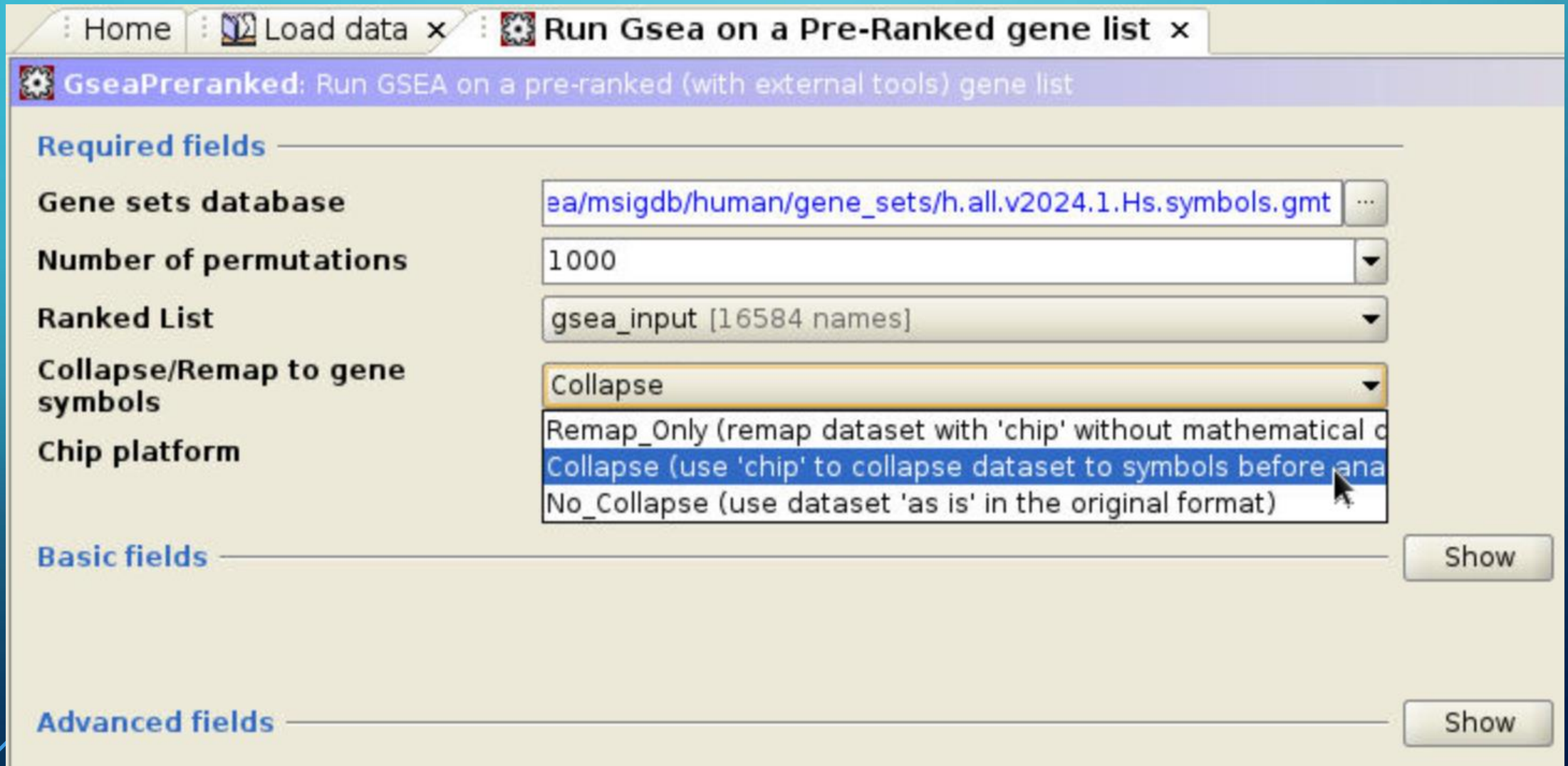


The screenshot shows the GSEA web interface with the following fields and options:

- Home** | **Load data** x | **Run Gsea on a Pre-Ranked gene list** x
- GseaPreranked:** Run GSEA on a pre-ranked (with external tools) gene list
- Required fields**
 - Gene sets database:** ea/msigdb/human/gene_sets/h.all.v2024.1.Hs.symbols.gmt
 - Number of permutations:** 1000
 - Ranked List:** gsea_input [16584 names]
 - Collapse/Remap to gene symbols:** Remap_Only (highlighted with a red box and a red arrow pointing to the drop-down arrow)
 - Chip platform:**
- Basic fields** **Show**
- Advanced fields** **Show**

RECREATE FIGURES: FIG. 4

Select **Collapse**:



Home Load data x Run Gsea on a Pre-Ranked gene list x

GseaPreranked: Run GSEA on a pre-ranked (with external tools) gene list

Required fields

Gene sets database

Number of permutations

Ranked List

Collapse/Remap to gene symbols

Chip platform

Basic fields

Advanced fields

Remap_Only (remap dataset with 'chip' without mathematical d
Collapse (use 'chip' to collapse dataset to symbols before ana
No_Collapse (use dataset 'as is' in the original format)

RECREATE FIGURES: FIG. 4

Click on the button with three dots (...) for the “**Chip platform**”:



Home Load data x Run Gsea on a Pre-Ranked gene list x

GseaPreranked: Run GSEA on a pre-ranked (with external tools) gene list

Required fields

Gene sets database ...

Number of permutations

Ranked List

Collapse/Remap to gene symbols

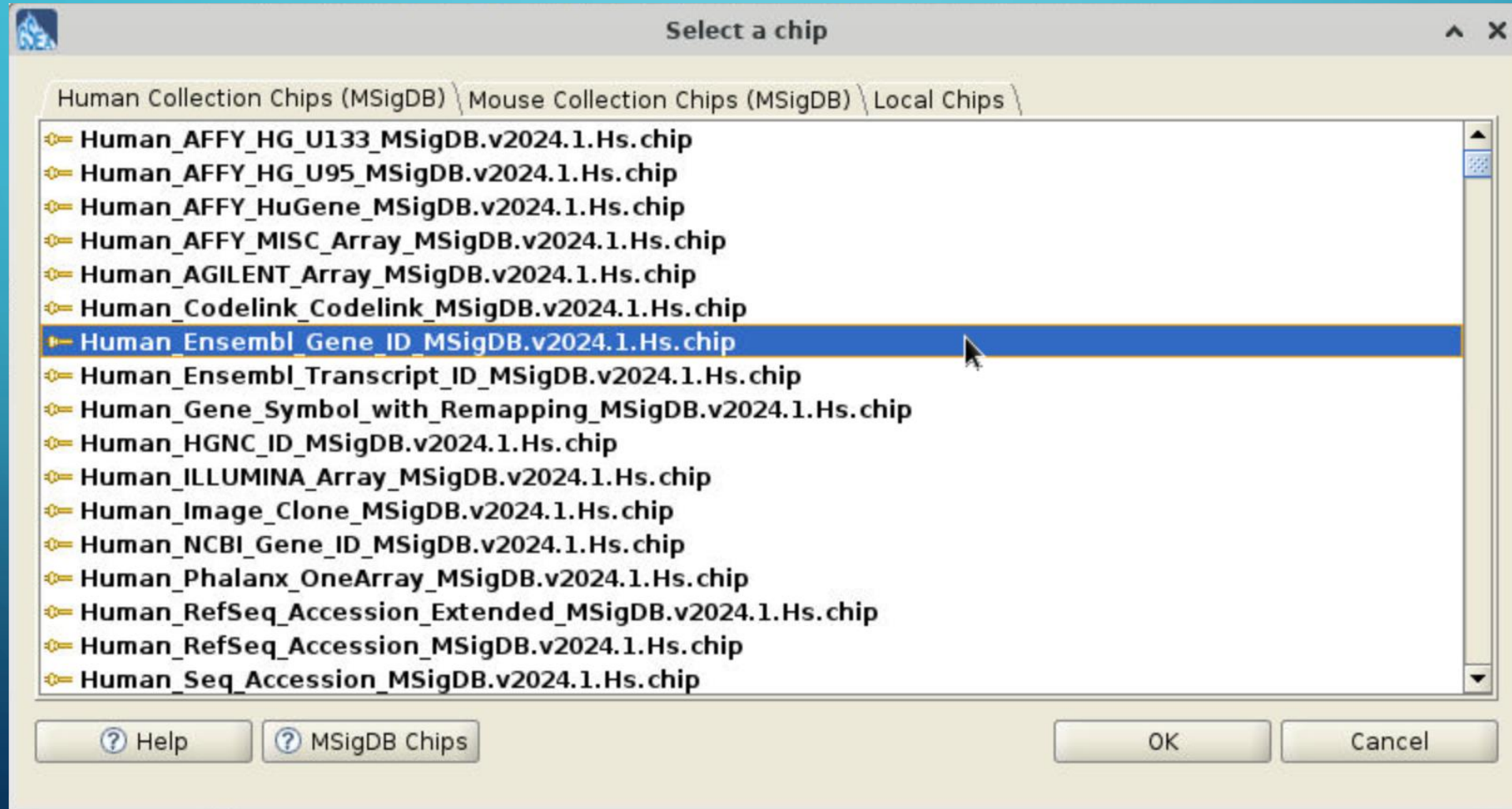
Chip platform ...

Basic fields

Advanced fields

RECREATE FIGURES: FIG. 4

Select (double-click) the **7th set**, “**Human_Ensembl_Gene_ID_MSigDB.v2024.1.Hs.chip**”



RECREATE FIGURES: FIG. 4

Click the “**Run**” button on the bottom:

 **GseaPreranked:** Run GSEA on a pre-ranked (with external tools) gene list

Required fields

Gene sets database	ea/msigdb/human/gene_sets/h.all.v2024.1.Hs.symbols.gmt ...
Number of permutations	1000 ▼
Ranked List	gsea_input [16584 names] ▼
Collapse/Remap to gene symbols	Remap_Only ▼
Chip platform	tations/Human_Ensembl_Gene_ID_MSigDB.v2024.1.Hs.chip ...

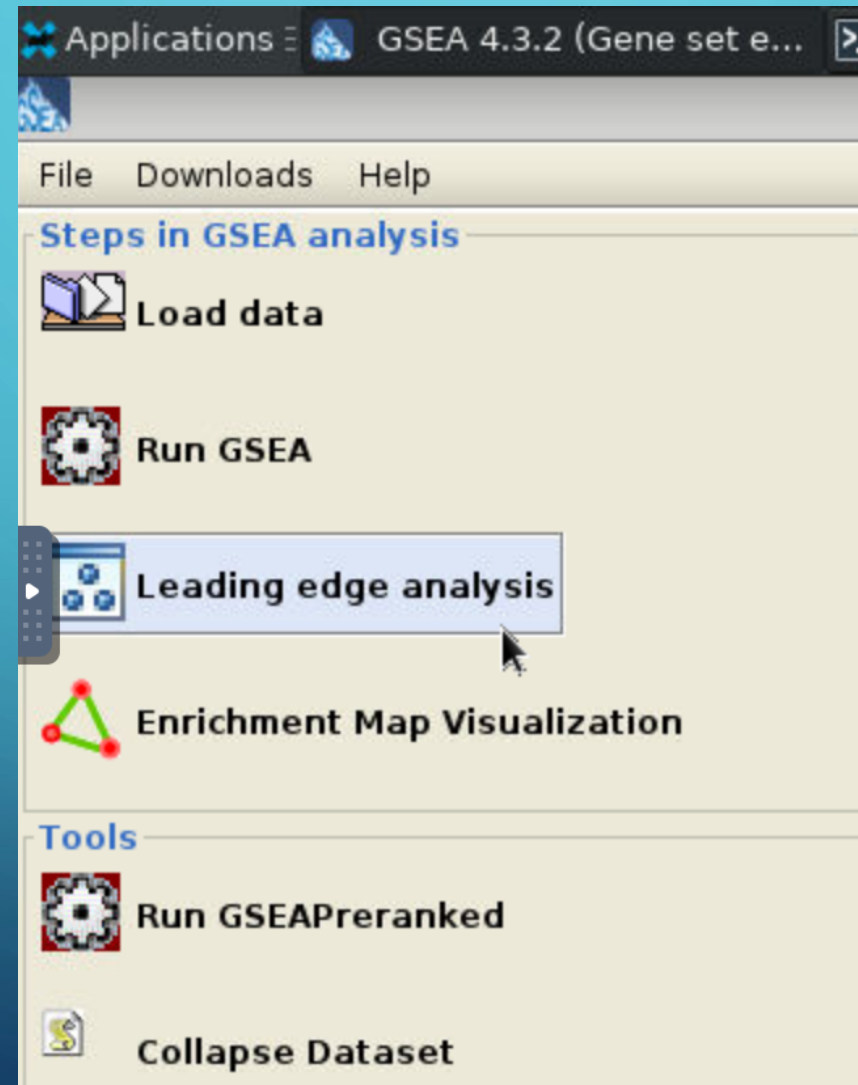
Basic fields

Advanced fields

ⓘ Reset ⚡ Last 📄 Command ▶ **Run**

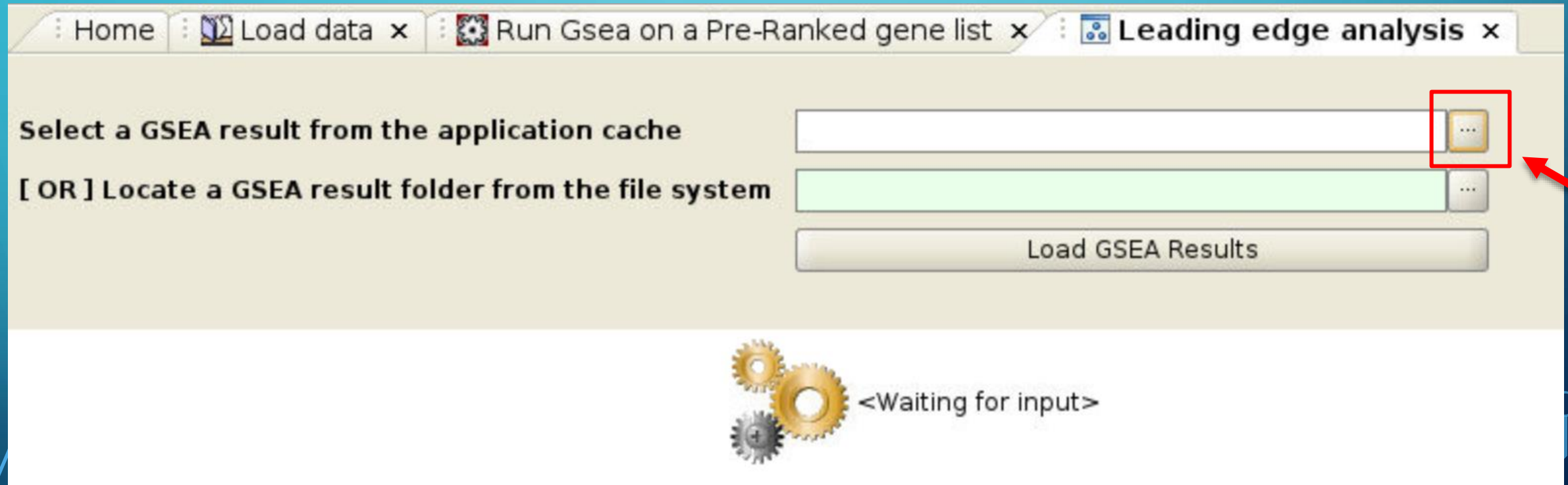
RECREATE FIGURES: FIG. 4

Click the “**Leading edge analysis**” in the menu on the left:



RECREATE FIGURES: FIG. 4

Click on the button with three dots (...) for the **Select a GSEA result from the application cache**:



Home Load data x Run Gsea on a Pre-Ranked gene list x Leading edge analysis x

Select a GSEA result from the application cache

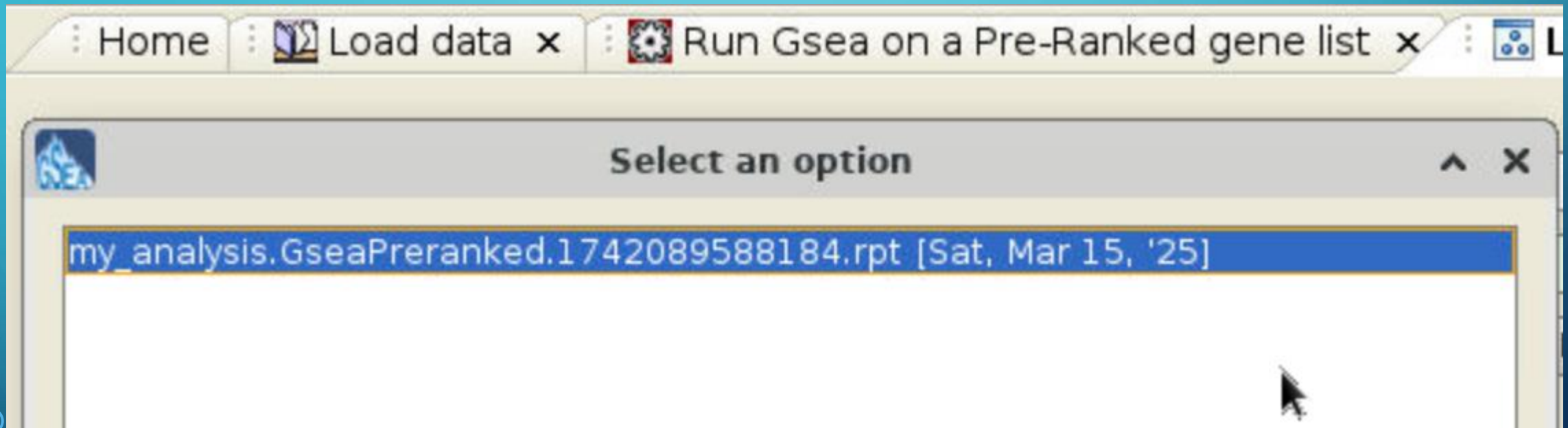
[OR] Locate a GSEA result folder from the file system

Load GSEA Results

<Waiting for input>

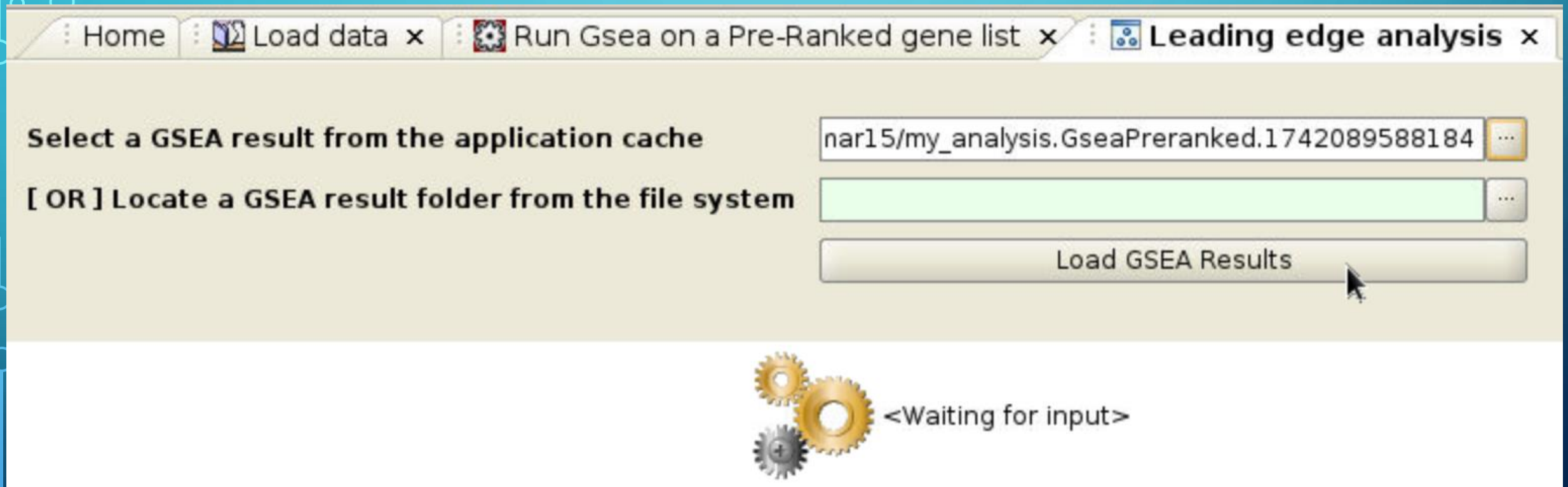
RECREATE FIGURES: FIG. 4

Select your results:



RECREATE FIGURES: FIG. 4

Click the “**Load GSEA Results**” button:



The screenshot shows the GSEA web application interface. The top navigation bar includes tabs for 'Home', 'Load data', 'Run Gsea on a Pre-Ranked gene list', and 'Leading edge analysis'. The 'Leading edge analysis' tab is active. Below the tabs, there are two input fields for selecting a GSEA result. The first field is labeled 'Select a GSEA result from the application cache' and contains the text 'nar15/my_analysis.GseaPreranked.1742089588184'. The second field is labeled '[OR] Locate a GSEA result folder from the file system' and is empty. Below these fields is a large button labeled 'Load GSEA Results'. A mouse cursor is hovering over the button. At the bottom of the interface, there is a gear icon and the text '<Waiting for input>'. The background of the interface is light blue with a circuit-like pattern.

Home Load data x Run Gsea on a Pre-Ranked gene list x Leading edge analysis x

Select a GSEA result from the application cache nar15/my_analysis.GseaPreranked.1742089588184 ...

[OR] Locate a GSEA result folder from the file system ...

Load GSEA Results

<Waiting for input>

RECREATE FIGURES: FIG. 4

Sort (ascending) the results by “**FDR q-val**” by clicking on the **FDR q-val** tab:

positive phenotype: na pos negative phenotype: na neg

Filter Gene Sets

Q- 50 out of 50 gene sets

Gene Set	Size	ES	NES	NOM p-val	FDR q-val	FWER p-val	Rank at Max	Leading Edge
HALLMARK_UN...	112	0.731	2.876	0	0	0	695 tags=38%, lis...	
HALLMARK_IN...	78	0.7	2.606	0	0	0	2,557 tags=60%, lis...	
HALLMARK_PR...	93	0.638	2.45	0	0	0	2,298 tags=41%, lis...	
HALLMARK_IN...	141	0.571	2.323	0	0	0	2,932 tags=50%, lis...	
HALLMARK_E2...	199	-0.44	-2.1	0	0	0	2,665 tags=40%, lis...	
HALLMARK_ES...	192	-0.454	-2.168	0	0	0	2,590 tags=40%, lis...	
HALLMARK_MY...	200	-0.499	-2.392	0	0	0	4,497 tags=58%, lis...	
HALLMARK_MY...	58	-0.779	-3.005	0	0	0	2,009 tags=74%, lis...	
HALLMARK_MT...	197	0.501	2.154	0	0	0.002	1,240 tags=24%, lis...	
HALLMARK_TN...	164	0.502	2.09	0	0.001	0.008	1,799 tags=35%, lis...	
HALLMARK_P5...	184	0.459	1.938	0	0.003	0.024	2,678 tags=33%, lis...	
HALLMARK_HY...	172	0.451	1.882	0	0.004	0.032	1,757 tags=24%, lis...	
HALLMARK_CH...	69	0.518	1.892	0.003	0.004	0.032	3,299 tags=43%, lis...	
HALLMARK_ES	181	-0.364	-1.714	0	0.005	0.025	3,067 tags=35%, lis...	

For 0 selected gene sets:

RECREATE FIGURES: FIG. 4

Highlight all the rows up to FDR q-val < 0.25:

positive phenotype: na pos negative phenotype: na neg

Filter Gene Sets

50 out of 50 gene sets

Gene Set	Size	ES	NES	NOM p-val	FDR q-val ▲	FWER p-val	Rank at Max	Leading Edge
HALLMARK_HE...	157	0.351	1.439	0.026	0.077	0.657	2,144	tags=25%, lis...
HALLMARK_IL6...	53	0.403	1.404	0.051	0.085	0.737	2,008	tags=28%, lis...
HALLMARK_PL...	92	0.356	1.371	0.046	0.101	0.823	2,152	tags=24%, lis...
HALLMARK_PA...	16	0.511	1.352	0.122	0.102	0.859	3,313	tags=31%, lis...
HALLMARK_MI...	196	0.317	1.358	0.028	0.103	0.847	2,488	tags=29%, lis...
HALLMARK_AP...	31	-0.358	-1.2	0.189	0.173	0.9	934	tags=19%, lis...
HALLMARK_AL...	96	0.319	1.227	0.141	0.22	0.986	2,306	tags=27%, lis...
HALLMARK_KR...	105	0.281	1.103	0.271	0.436	1	3,857	tags=33%, lis...
HALLMARK_ALLOGRAFT_REJECTION		0.266	1.081	0.296	0.469	1	1,800	tags=12%, lis...
HALLMARK_FA...	126	0.259	1.061	0.321	0.478	1	2,688	tags=20%, lis...

RECREATE FIGURES: FIG. 4

Click the **“Run leading edge analysis”** button on the bottom right:

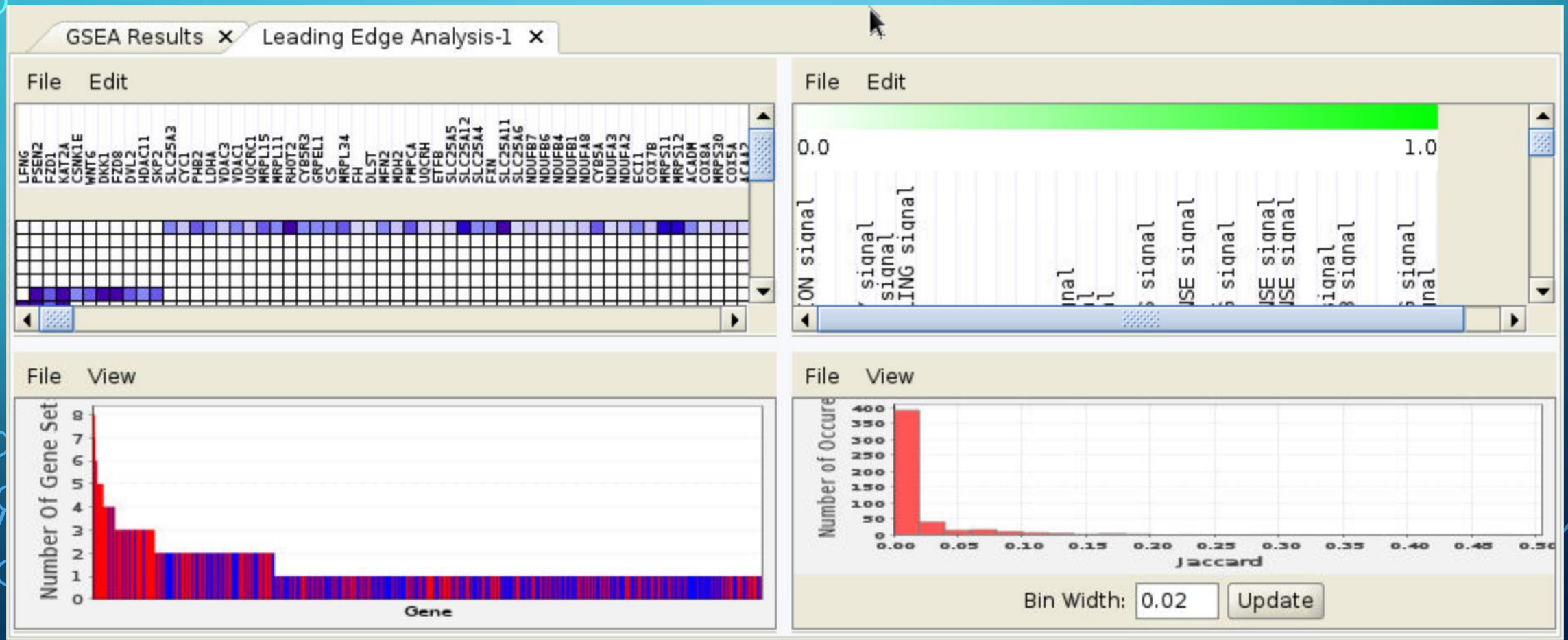
HALLMARK_PA...	16	0.511	1.359	0.096	0.102	0.831	3,313	tags=31%, lis...
HALLMARK_MI...	196	0.317	1.355	0.026	0.1	0.838	2,488	tags=29%, lis...
HALLMARK_AL...	96	0.319	1.213	0.148	0.244	0.989	2,306	tags=27%, lis...
HALLMARK_KR...	105	0.281	1.103	0.263	0.439	1	3,857	tags=33%, lis...
HALLMARK_XE...	138	0.266	1.085	0.289	0.46	1	1,800	tags=12%, lis...
HALLMARK_RE...	46	0.317	1.082	0.32	0.445	1	1,850	tags=15%, lis...

For 20 selected gene sets:

rting: RankedList from: /home/databiom/gsea_home/output/mar15/my_analysis.GseaPreranked.1742089588184/edb/g...
 622M of 2048M

RECREATE FIGURES: FIG. 4

Leading Edge Results:



RECREATE FIGURES: FIG. 4

Go back to **GSEA Results** by clicking on the tab. Then click **Build HTML Report**:

GSEA Results x Leading Edge Analysis-1 x

positive phenotype: na pos negative phenotype: na neg

Filter Gene Sets

Q- 50 out of 50 gene sets

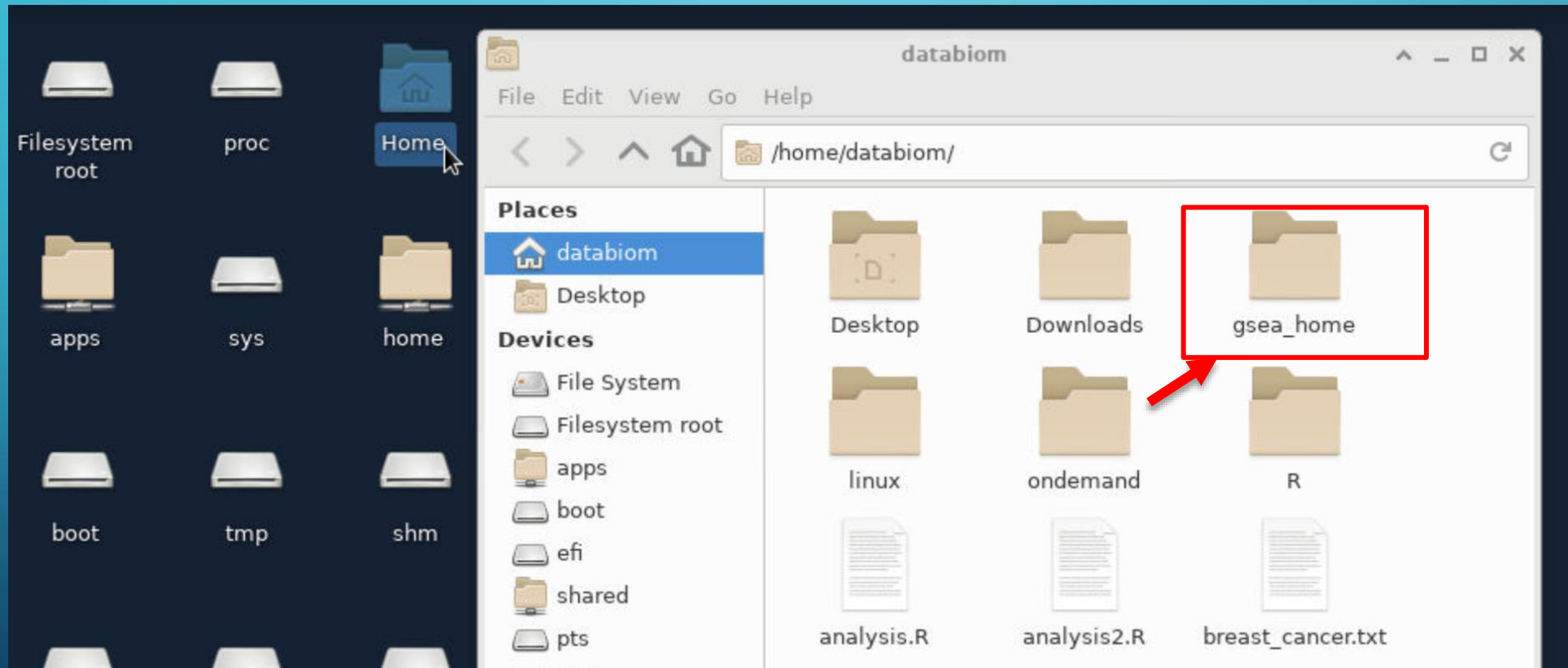
Gene Set	Size	ES	NES	NOM p-val	FDR q-val	FWER p-val	Rank at Max	Leading Edge
HALLMARK_IN...	104	0.442	1.752	0.004	0.011	0.099	3,110	tags=40%, lis...
HALLMARK_AP...	126	0.405	1.626	0.004	0.026	0.244	2,253	tags=29%, lis...
HALLMARK_CO...	131	0.387	1.556	0.01	0.039	0.364	2,995	tags=34%, lis...
HALLMARK_AN...	93	0.395	1.495	0.02	0.054	0.49	866	tags=14%, lis...
HALLMARK_GL...	179	0.343	1.467	0.013	0.062	0.574	1,298	tags=17%, lis...
HALLMARK_HE...	157	0.351	1.451	0.02	0.065	0.607	2,144	tags=25%, lis...
HALLMARK_IL6...	53	0.403	1.397	0.062	0.088	0.734	2,008	tags=28%, lis...
HALLMARK_PI...	92	0.356	1.382	0.055	0.092	0.774	2,152	tags=24%, lis...
HALLMARK_PA...	16	0.511	1.359	0.096	0.102	0.831	3,313	tags=31%, lis...
HALLMARK_MI...	196	0.317	1.355	0.026	0.1	0.838	2,488	tags=29%, lis...
HALLMARK_AL...	96	0.319	1.213	0.148	0.244	0.989	2,306	tags=27%, lis...
HALLMARK_KR...	105	0.281	1.103	0.263	0.439	1	3,857	tags=33%, lis...

For 20 selected gene sets:

Run leading edge analysis Build HTML Report

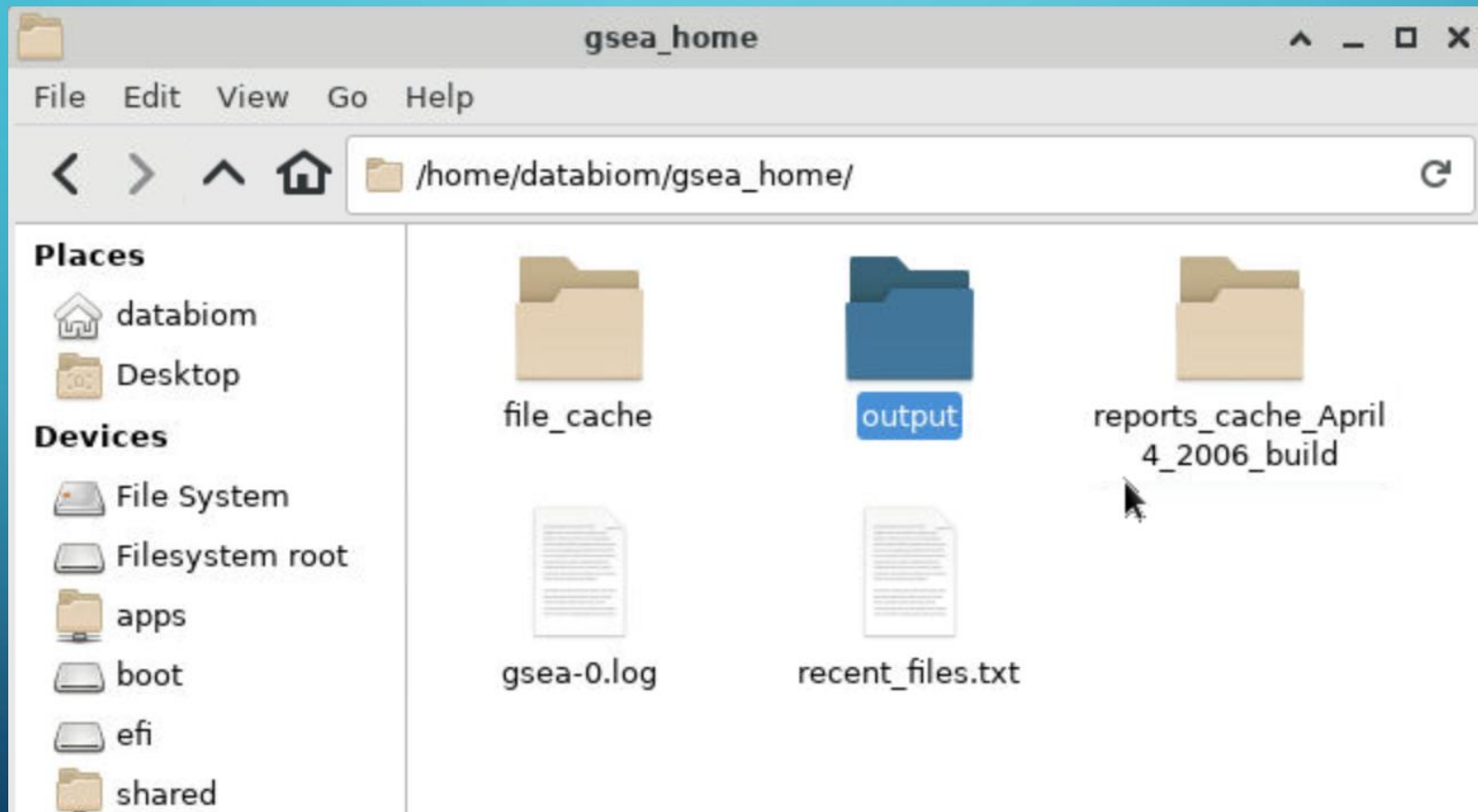
RECREATE FIGURES: FIG. 4

Close GSEA and the terminal. Navigate to your **home folder** and open it. Double-click the “**gsea_home**” folder.



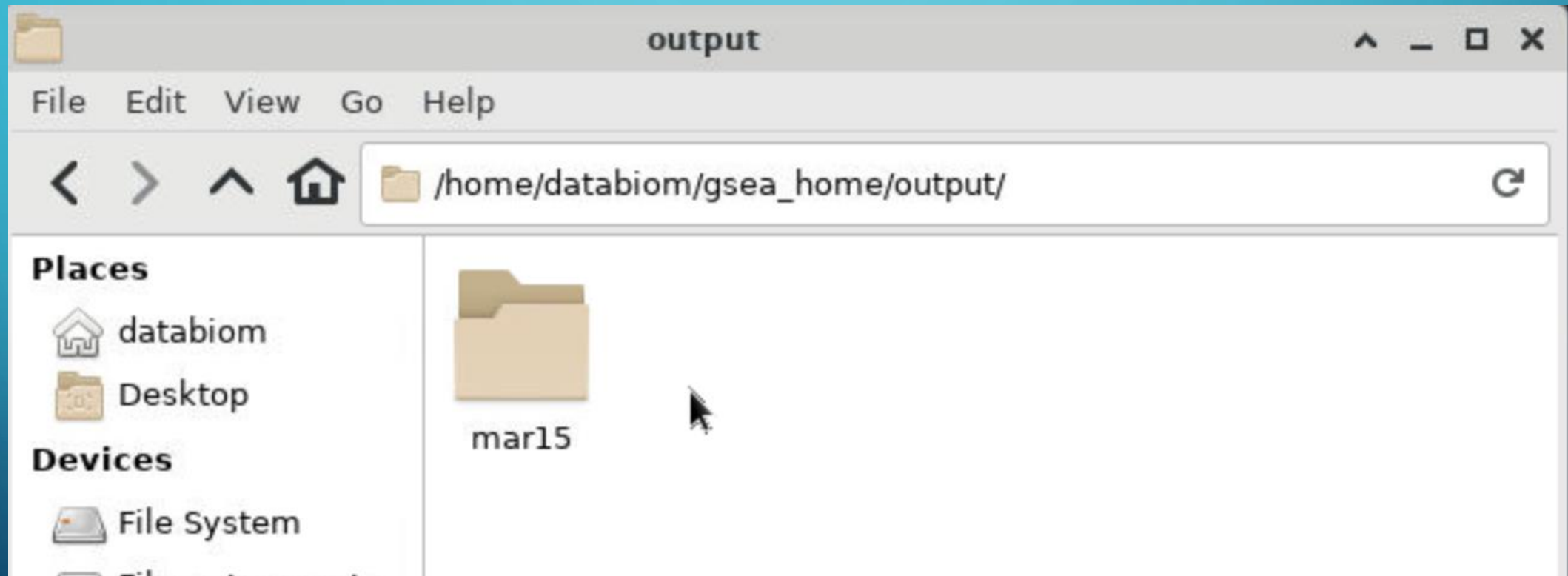
RECREATE FIGURES: FIG. 4

Open the “**output**” folder:



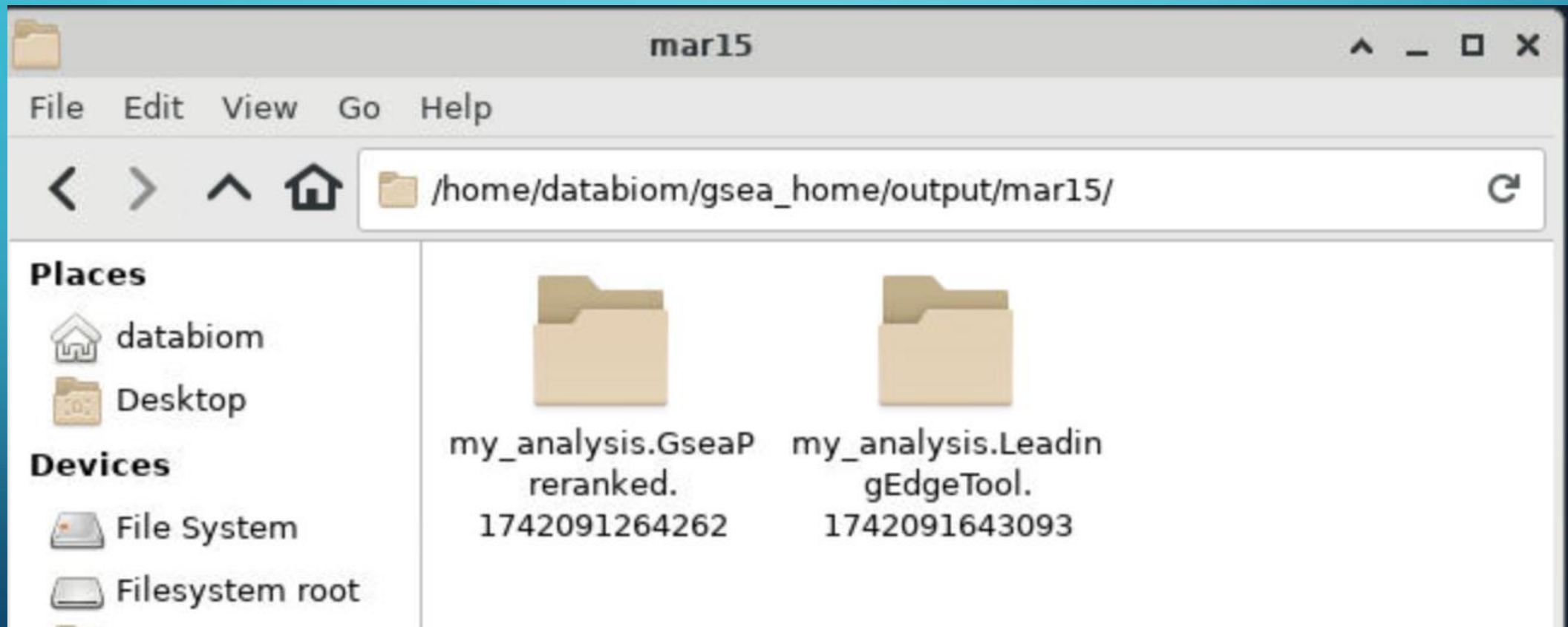
RECREATE FIGURES: FIG. 4

Open the dated folder:



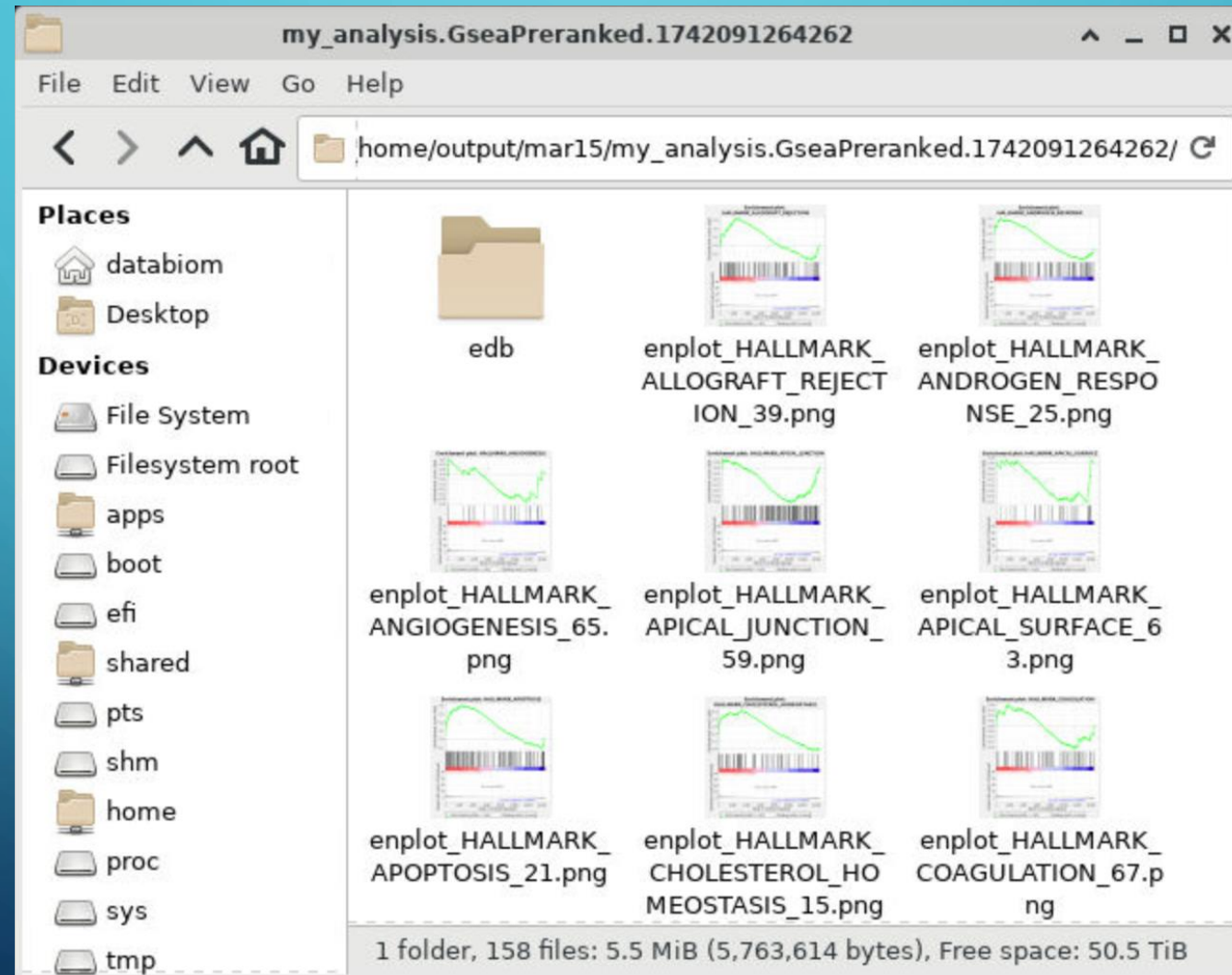
RECREATE FIGURES: FIG. 4

Within these folders are the results (plots, reports, etc.)



RECREATE FIGURES: FIG. 4

GSEA enrichment plots:



GOOD JOB!

This concludes the bulk RNA-seq tutorial.



Don't forget to save your script if you made any edits.

